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Agricultural Diversity, Structural Change, and Long-Run Development: Evidence from the United States†

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This paper examines the role of agricultural diversity in the process of development. Using data from US counties and exploiting climate-induced variation in agricultural production patterns, I show that mid-nineteenth-century agricultural diversity had positive long-run effects on population density and income per capita. During the Second Industrial Revolution, agricultural diversity fostered industrialization, diversification within manufacturing, patent activity, formation of new labor skills, and the expansion of knowledge- and skill-intensive industries. These results are consistent with the hypothesis that diversity spurs the acquisition of new ideas and new skills because of the presence of cross-sector spillovers and complementarities. (*JEL* N31, N32, N51, N52, N91, N92, Q10)

At stages of development when agriculture employs a large share of the population, the characteristics of agricultural production can strongly influence the evolution of the economy. Economists have extensively studied the impact of agricultural productivity on development (e.g., Restuccia, Yang, and Zhu 2008; Gollin 2010; Hornbeck and Keskin 2015; Bustos, Caprettini, and Ponticelli 2016). The effects of certain types of agricultural products have also attracted considerable interest (e.g., Goldin and Sokoloff 1984; Engerman and Sokoloff 2002; Nunn and Qian 2011; Vollrath 2011; Bruhn and Gallego 2012). In contrast, the role of diversity (the variety and balance of the production mix) in agriculture has remained largely unexplored. This paper shows that agricultural diversity can be an important driver of structural change and economic growth.

To examine how agricultural diversity affected the process of development, I use data from US counties over 140 years and propose a novel instrumental variable (IV) strategy. I find significant positive effects of mid-nineteenth-century agricultural diversity on development following the onset of the Second Industrial Revolution

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(usually dated 1870–1920). Early agricultural diversity—measured in 1860, when the American economy was still predominantly agricultural—did not have a significant impact on contemporaneous development across US counties. But in the decades that followed, while the nation was becoming the world's industrial leader, county-level industrialization and population density were positively affected by early agricultural diversity. The effects were persistent and sizable: according to my IV estimates, a 1 standard deviation increase in 1860 agricultural diversity led to an increase of 73 percent in 2000 population density and a gain of 6 percent in 2000 per capita income.

In theory, production diversity may affect development in either direction, through various channels. Diversity can hinder development, as it may imply foregoing gains from specialization based on comparative advantage or scale economies. On the other hand, if there are complementarities among inputs or skills, diversity may yield productivity gains. Moreover, diverse economic environments may favor technological change and skill formation. Diversity could also foster development through other channels, e.g., by reducing risk and volatility. Empirically identifying the causal effects of diversity is challenging: while diversity may affect development in various ways, the latter may also affect the former, and there are third factors that may affect both. In this paper, I address the challenge exploiting climate-induced variation in agricultural production patterns and assess the plausibility of different mechanisms drawing on a wealth of historical data.

I start with a preliminary exploration of the relationship between early agricultural diversity and development outcomes through ordinary least squares (OLS) regressions. I find significant positive correlations that are robust to controlling for state fixed effects, land productivity measures and several other geoclimatic features, the dominance of cotton and other specific agricultural products, and a number of socioeconomic initial conditions. But despite their robustness, the correlations could be driven by heterogeneity in unobservables (e.g., preferences or technology). For instance, if diversity in early agriculture partly reflected openness to new ideas and this trait accelerated subsequent growth, OLS estimates would be biased upward. A downward bias generated by other omitted variables is also possible.

I propose an identification strategy that exploits variation in the composition of agricultural production generated by climatic features. I construct an IV for agricultural diversity using high-resolution spatial data from FAO (the Food and Agriculture Organization) on climate-based potential yields for many different crops. The IV is based on the estimation of a fractional multinomial logit (FML) model of crop choice, in which the county-level shares of agricultural products in agricultural output are functions of the crop-specific potential yields. With the predicted shares for each crop, I compute an index of potential agricultural diversity, which I then use as an IV for actual diversity.

The findings are robust to controlling flexibly for several measures of overall land productivity and crop-specific potential yields; latitude and longitude; temperature and rainfall; and distances to rivers, canals, and the fall line. I also consider various ways of controlling for the importance of specific crops and add a number of socioeconomic controls that account more thoroughly for initial conditions. The estimates remain significant and relatively stable throughout this battery of checks, mitigating concerns about the exclusion restriction in the IV estimation.

After establishing the positive effects of agricultural diversity on development, I investigate the underlying mechanisms. My proposed explanation emphasizes that agricultural diversity can foster industrial diversification and the acquisition of new ideas and new skills. To motivate the hypothesis in the context of the Second Industrial Revolution, I offer a collection of examples that illustrate the ramifications of early agricultural diversity in US history. Agricultural products required different skills and technologies in production, storage, packaging, marketing, and transportation. In addition, they had linkages with various manufacturing activities, which in turn required different technologies and skills, and had a variety of other linkages. Thus, directly and indirectly, agricultural diversity may have increased the local diversity of products, ideas, and skills. In turn, this diversity may have favored technological change and skill formation because of the presence of complementarities and cross-sector spillovers.[1]

To assess this hypothesis, I estimate the impacts of agricultural diversity on industrial diversity, patent activity, and the share of manufacturing employment in new occupations. The results show that these key intermediate variables, measured in the decades following the onset of the Second Industrial Revolution, were all positively affected by early agricultural diversity. The proposed mechanisms have an additional testable implication: the effects of diversity ought to be larger in activities where skill complementarities and cross-sector knowledge spillovers are more relevant. Consistent with this prediction, I find that beyond its effect on overall industrialization, agricultural diversity positively affected the shares of manufacturing workers employed in knowledge- and skill-intensive industries.

There are other channels, unrelated to the cross-sector spillovers and complementarities emphasized above, that could account for the positive effects of agricultural diversity on development: changes in agricultural productivity (which could in turn boost industrialization), reduced exposure to product-specific shocks (which would decrease volatility and possibly foster economic performance), or lower land concentration (which could affect local institutions, e.g., schools and banks). I assess the relevance of these channels and do not find evidence supporting any of them.

My findings contribute to a growing body of knowledge on the deep roots of comparative development (see, e.g., Nunn 2014; Ashraf and Galor 2017), offering novel insights on how the production structure at early stages of development can affect long-run performance. Like the recent contributions of Bleakley and Lin (2012); Hornbeck and Naidu (2014); and Glaeser, Kerr, and Kerr (2015), I find evidence from US history indicating significant long-run effects of local conditions that are no longer directly relevant. Among the many contributions that study how geoclimatic factors have affected contemporary outcomes by shaping historical development paths, my research closely echoes those highlighting the role of early agricultural production patterns (e.g., Engerman and Sokoloff 2002). But the focus

[1] In online Appendix B, I present a formal model of these mechanisms that emphasizes the interconnected roles of entry into new activities and the acquisition of ideas and skills in the process of economic growth.

on agricultural diversity, instead of agricultural productivity or specific crops, provides a new perspective on the role of agriculture in the development process.[2]

This paper also adds to the urban and regional economics literature. The mechanisms that I emphasize echo the ideas of Jacobs (1969), which were quantitatively studied in the seminal contribution of Glaeser et al. (1992) and many follow-up studies. While many of these studies present suggestive evidence about the relationship between diversity and development, a causal interpretation is generally subject to endogeneity concerns. In this paper, the focus on agriculture offers a useful lever for identification, as it allows me to isolate climate-induced variation in diversity.

Finally, my results are relevant for the macro-development literature. My estimation of diversity's effects and the emphasis on skills formation and technological change as the underlying channels (along the lines of Hausmann and Hidalgo 2011) complement the influential contributions of Acemoglu and Zilibotti (1997); Imbs and Wacziarg (2003); and Koren and Tenreyro (2013), which link diversity and development in different ways. Moreover, my results and interpretation have implications about the mechanics of structural change: they suggest that analyzing complementarities and cross-sector spillovers (e.g., along the lines of Jones 2011; Hanlon and Miscio 2017; or Cai and Li 2019) in models with finer levels of aggregation than standard two- or three-sector frameworks may produce novel insights.

The paper is organized as follows. Section I discusses the theories that link diversity and development. Section II describes the data and historical background, and Section III presents the estimating equation and exploratory results from OLS estimation. Section IV outlines the instrumental variable strategy and discusses the IV estimates. Section V investigates the mechanisms underlying the long-term effects of agricultural diversity. Section VI concludes.

I. Theories Linking Economic Diversity and Development

The relationship between economic diversity and development has been addressed by a number of theories with contrasting implications about the sign and direction of causality. This section briefly describes this multiplicity of theories and elaborates on those that generate positive effects of diversity because of the presence of complementarities and knowledge spillovers. This subset of theories informs my interpretation of how diversity in early agricultural production affected the process of development in US history, as explained in detail throughout the paper.

Well-known pieces of economic wisdom suggest that specialization, as opposed to diversification, is good for growth. First, any form of increasing returns to scale operating only at the sector or product level implies that specialization yields productivity gains; within urban economics, for instance, the idea of localization externalities points to the benefits that firms derive from being co-located with other firms in the same sector. In addition, the principle of comparative advantage is often

[2] While many important contributions study diversity along other dimensions (e.g., ethnic, cultural, or genetic diversity), the closest precedents to my focus on agricultural diversity are Michalopoulos (2012) and Fenske (2014), who examine the effects of different measures of diversity in natural endowments on ethnolinguistic fragmentation and state centralization, respectively (both in the context of Africa).

used to make a case for specialization. Finally, an elementary consideration of modern portfolio theory may suggest that growth is diminished by diversification insofar as (with the purpose of attenuating risk) it reduces expected returns.

In contrast, several theories imply positive effects of diversity. A number of different macro-development models feature complementarities (i.e., imperfect substitutability) among inputs in the production function—intermediate goods, skills, tasks, or technologies, depending on the case—with the implication that diversity increases productivity (in the sense of lowering unit costs). Some models (e.g., endogenous growth models with expanding variety a la Romer 1990) highlight the productivity gains from adding new production inputs. Others focus on the role of balance among required inputs—e.g., Jones (2011), who sheds light on the importance of "weak links" in the presence of strong complementarities.

There are also theories that generate *dynamic* advantages of diversity. Jacobs (1969) eloquently argued that diverse economic environments promote the acquisition of new ideas and skills. These effects may be explained by cross-product knowledge spillovers, particularly by recombinant technological progress—the production of new ideas by combining multiple old ideas (see, e.g., Weitzman 1998; Olsson 2005; Berliant and Fujita 2011). Zeppini and van den Bergh (2013) show that even if there are increasing returns that favor specialization, diversity's stimulus to recombinant innovations can make it efficient in the long run.

Other models that generate dynamic advantages emphasize diversity in inputs and skills. Duranton and Puga (2001) build a model in which diversified cities, characterized by a broad range of intermediate inputs and labor skills, allow an entrepreneur with a new project to find the ideal productive process. In a model developed by Helsley and Strange (2002), the diversity of input suppliers reduces the cost of bringing new ideas to fruition. Hausmann and Hidalgo (2011) emphasize diversity in skills, or more broadly, capabilities: if each product requires a set of complementary capabilities, and capability sets for different products partially overlap, then diversity increases the return to acquiring new capabilities, insofar as they complement previously existing ones.

Various elements of the theories described above are closely related and can be jointly incorporated in a stylized model, as I do in online Appendix B. The model features skill complementarities and cross-sectoral spillovers and generates positive effects of diversity on both the acquisition of new ideas and the formation of new skills. Each sector requires a number of complementary skills. The expected level of efficiency for each skill positively depends—through spillovers—on the variety of established skills in other sectors of the local economy. Higher diversity reflects a wider range of skills, which increases expected returns and thus entry into new sectors. In turn, expanding the set of active industrial sectors goes along with the adoption of new technologies and the formation of new skills. This subsequently fosters entry into additional activities, boosting the process of growth and structural change.

Positive effects of diversity could also arise from reduced risk and volatility. Diversification limits the direct impact of negative product-specific shocks and also facilitates substitution away from negatively affected products. Moreover, if agents are risk averse, being able to enter a wide array of projects may be a precondition for carrying out risky projects with high returns. These insights are suggested by

Acemoglu and Zilibotti (1997) and Koren and Tenreyro (2013), two important contributions characterizing the endogenous evolution of diversification and volatility in the process of development.

A positive relationship between diversity and development might also be driven, under some special conditions, by trade integration. While the standard theories predict that trade induces specialization and also increases income, Imbs, Montenegro, and Wacziarg (2012) show that at low income levels and under indivisibilities in some sectors, enlarging market access can make minimum scales feasible and thus induce entry into new sectors, which would increase diversity and also income.[3]

Finally, diversity and development might be connected by an effect of the latter on the former: Li (2021) shows that higher income can lead to increased variety in consumption (which may in turn induce higher diversity in production).

In sum, there are multiple ways in which diversity can affect development, as well as ways in which development might affect diversity or a third variable might affect both. The theories reviewed in this section do not have a focus on *agricultural* diversity, but their insights remain relevant in the context studied in this paper. In 1860, agriculture was still the main sector of the US economy. In addition, the mechanisms that I emphasize may have been activated by agricultural diversity directly, or indirectly, through industrial diversity induced by early agricultural diversity. The empirical analysis in Section V and the model in online Appendix B provide a precise account of my interpretation of agricultural diversity's effects on development in US history.

II. Data and Historical Background

The empirical analysis is based on a sample of 1,821 US counties that represented over 97 percent of the national population in 1860. The sample excludes all Western states (most of which were still territories by 1860 and had only started to be partitioned into counties) as well as Dakota Territory, Kansas Territory, Nebraska Territory, and Indian Territory (later Oklahoma).[4] To have consistent units of observation over time in spite of changes in county boundaries, I adjust all data to conform to 1860 boundaries; the procedure is explained in the Data Appendix.

To measure early agricultural diversity, I use data on production of 36 items comprising agricultural output from the 1860 Census of Agriculture, in combination with data on state-level prices from Atack and Bateman (1987) and Craig (1993) (both available from Manson et al. 2019). Table 1 reports the sample average and maximum percentages of county-level agricultural production for each of these products as well as the percentages of counties in which each product was dominant and

[3] Imbs, Montenegro, and Wacziarg (2012) put together this prediction and the usual one to show that trade integration occurring first between regions (at relatively low income levels) and subsequently between countries (at higher income levels) can generate a country-level inverted U shape of diversification in relation to income (Imbs, Montenegro, and Wacziarg 2012, and the earlier contribution by Imbs and Wacziarg 2003, empirically document this pattern with data from 1960 onward).

[4] Within the remaining states, 85 counties are dropped because of missing data, many of them in Michigan and Texas. The sample includes El Paso, Texas, which is about 400 miles away from the closest county in the sample; excluding it does not qualitatively change the results. For convenience, El Paso is omitted from the maps (Figures 1, 3, and 6).

Table 1—Agricultural Production Data

	Percent of agriculture output		Percent of counties	
Product	Mean	Max	Dominant	>50%
Corn	23.80	98.89	43.66	11.26
Ginned cotton	16.03	94.11	20.04	13.84
Animals slaughtered	13.07	95.16	4.34	0.55
Hay	11.86	73.97	17.24	1.65
Wheat	10.39	92.86	8.40	0.44
Butter	4.22	25.36	0.05	0.00
Oats	3.92	23.83	0.05	0.00
Irish potatoes	3.10	59.87	0.77	0.11
Tobacco	2.34	57.27	2.58	0.16
Sweet potatoes	1.26	30.15	0.22	0.00
Orchards	1.18	24.18	0.00	0.00
Cane sugar	1.15	88.43	0.77	0.66
Wool	1.01	100.00	0.22	0.22
Rye	0.92	15.55	0.00	0.00
Market gardens	0.91	98.14	0.77	0.22
Peas and beans	0.72	22.95	0.00	0.00
Cheese	0.70	35.70	0.22	0.00
Buckwheat	0.56	15.31	0.00	0.00
Barley	0.44	10.54	0.00	0.00
Clover seed	0.30	12.93	0.00	0.00
Rice	0.27	86.38	0.50	0.27
Dew-rotted hemp	0.26	64.86	0.05	0.05
Cane molasses	0.25	19.58	0.00	0.00
Sorghum molasses	0.25	7.19	0.00	0.00
Honey	0.23	8.41	0.00	0.00
Maple sugar	0.22	36.26	0.11	0.00
Hops	0.17	19.51	0.00	0.00
Grass seed	0.16	9.39	0.00	0.00
Hemp (other)	0.09	22.04	0.00	0.00
Maple molasses	0.06	7.43	0.00	0.00
Flax	0.05	4.83	0.00	0.00
Flaxseed	0.04	3.35	0.00	0.00
Beeswax	0.02	1.03	0.00	0.00
Wine	0.02	4.72	0.00	0.00
Water-rotted hemp	0.02	7.17	0.00	0.00
Silk cocoon	0.01	2.86	0.00	0.00

Notes: Based on county-level production data from the 1860 Census of Agriculture and state-level price data from Atack and Bateman (1987) and Craig (1993). The statistics for the 36 products included in agricultural production are for the sample of 1,821 counties considered in this paper. The first and second columns indicate the average and the maximum share of each product in agricultural output (average shares are weighted by county-level total agricultural production value so that they coincide with shares in total agricultural production in the sample); the third and fourth columns indicate, for each product, the percentages of counties in which it had the largest share in agricultural output and in which it represented over 50 percent of agricultural output, respectively.

in which it represented over 50 percent of agricultural output (average shares are weighted by counties' agricultural output, so they coincide with shares in agricultural production for the sample as a whole).

There were 5 products (corn, cotton, animals slaughtered, hay, wheat) representing shares of agricultural output above 10 percent each, but almost all 36 products represented sizable percentages of agricultural output in at least one county (for instance, rice represented less than 0.3 percent in the sample but over 86 percent in Georgetown, South Carolina). This indicates that there was considerable variation in agricultural production patterns across counties, which allows me to identify the effects of diversity beyond the importance of specific crops.

Using these data, I calculate a measure of agricultural diversity as 1 minus a Hirschman-Herfindahl index of the shares of each product in gross agricultural output: $Ag.\ Diversity_c = 1 - \sum_i \theta_{ic}^2$, where θ_{ic} (with $i = 1, 2, \ldots, 36$) is the share of product i in county c's agricultural production (in value terms). Online Appendix A.1 considers a number of alternative measures of agricultural diversity. The results of the paper are not driven by the specific functional form of the Hirschman-Herfindahl index. Moreover, the results are robust to using measures that address potential concerns about mismeasurement in the agricultural production data (e.g., in the value of animal products and cotton, for reasons specific to the 1860 data) or about the use of gross output data rather than value added data.

Panel A of Figure 1 shows the spatial distribution of agricultural diversity in 1860: it was high in the Northeast and Midwest but also in the Southeastern seaboard, along the Appalachian Mountains, and in Northern Texas. Beyond regional disparities, there were sizable differences across counties in each state, so I can identify the effects of agricultural diversity relying solely on within-state variation.

In 1860, the agricultural sector employed over 55 percent of the labor force and produced around 45 percent of total output in the US economy (Weiss 1992). Within my sample of 1,821 counties, over 80 percent of the population lived in rural areas. Three out of four of these counties did not have any urban population. Rural areas had some industrial production, but the bulk of manufacturing employment was located in major urban centers. In my sample, over 50 percent of total manufacturing employment in 1860 was concentrated in 25 counties.

Between 1860 and 1920, the share of the US labor force employed in agriculture halved. The economy grew faster than ever before, both in absolute and per capita terms, making the United States the biggest and most productive economy in the world. The United States overtook the United Kingdom in terms of labor productivity in the 1890s, and subsequently widened the gap, as a result of a larger shift of labor away from agriculture (the sector with the lowest productivity) and faster productivity growth in nonagricultural activities (Broadberry and Irwin 2006).

The average county-level share of population in manufacturing in my sample more than doubled its 1860 value by 1900, but there was wide variation across counties in the advance of the industrialization process. Small local economies had an important role in the takeoff of this period (Meyer 1989; Page and Walker 1991). While the importance of large industrial urban centers in manufacturing peaked by 1870 (their subsequent growth was led by services), some counties with little or no manufacturing production in 1860 became thriving mid-sized industrial localities by the early twentieth century. Among the counties with zero manufacturing production in 1860 in the sample (about 11 percent of the 1,821 counties), one out of three had a share of population in manufacturing above the median value in 1900. Panel B of Figure 1 shows the spatial distribution of this measure of industrialization; here again, besides the sharp disparity between the North (particularly the Northeast) and the South, there was significant variation within states.

The period circa 1870–1920, sometimes called the Second Industrial Revolution, was characterized by a series of massive transformations in technologies, spanning

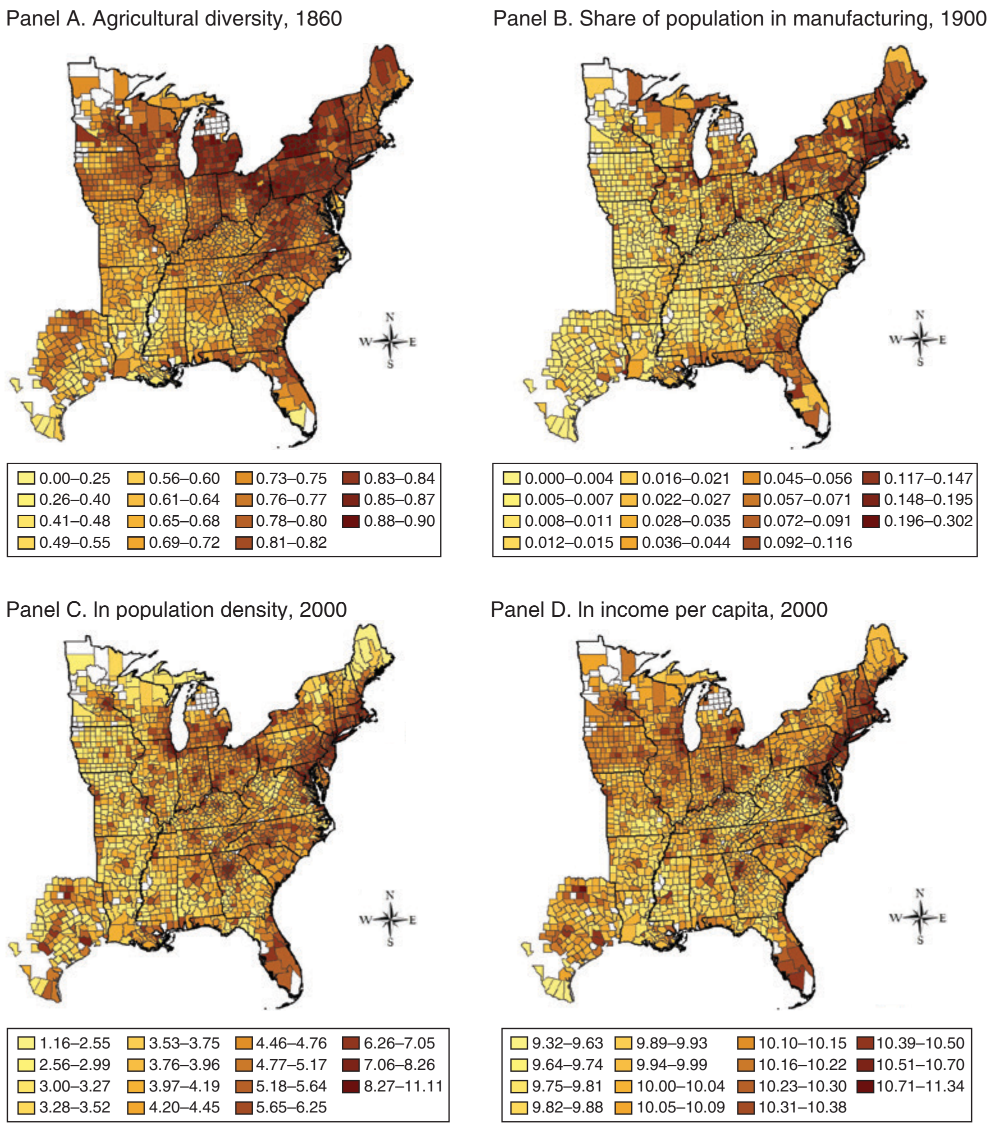

Figure 1. Agricultural Diversity, Industrialization, and Long-Run Development

Note: See Data Appendix for variable definitions and sources.

electricity, chemicals, steel, transportation, communications, management, and marketing. Electric power and continuous-process methods, two salient innovations of the time, were associated with the origins of the complementarity between technology and skill that would mark the twentieth century (Goldin and Katz 1998). The share of high-skill workers in the labor force increased, and there was a rapid expansion of primary and secondary education (Goldin and Katz 2010; Katz and Margo 2014).

The technological advances of the Second Industrial Revolution would continue to fuel growth in the following decades (David 1990; Gordon 2016). Innovations diffused progressively across different activities and types of firms as the growing supply of capital goods embodying the new technologies increased their profitability and as the acquisition of new skills and knowledge facilitated adoption. The effects on productivity were enhanced by subsequent incremental innovations based on learning-by-doing and by the introduction of new complementary technologies.

As measures of long-run development, I consider population density and income per capita in 2000 (in logs). Panel C and panel D of Figure 1 show their spatial distribution. Again, the highest levels (both for population density and income per capita) are concentrated in the Northeastern coast, but there is also considerable within-state variation. Note that while income per capita would be the natural outcome of interest in a context without labor flows across localities, population density may be the more meaningful outcome in a context of relatively high labor mobility, where income differentials (unless compensated by other factors) induce population flows.

Historical census data on manufacturing employment and population for 1860–1940 are drawn from Manson et al. (2019) and for later periods from County Data Books (Haines and ICPSR 2010). I use the share of the total population employed in the manufacturing sector as a measure of industrialization because labor force data is not consistently available before 1940.[5] Data on income per capita for 2000 come from the local area statistics of the Bureau of Economic Analysis (2015). Full details on variable definitions and sources are provided in the Data Appendix.

III. Estimating Equation and OLS Results

This section presents the estimating equation and a preliminary exploration of the relationship between early agricultural diversity and development through OLS estimation. An instrumental variable strategy to identify the effects of agricultural diversity is presented in the next section.

The estimating equation is

$$y_c = \alpha + \beta \, Ag.\ Diversity_{c,1860} + \delta_s + \gamma' \mathbf{X}_c + \varepsilon_c, \tag{1}$$

where y_c is a development outcome for county c (in the main regressions, income per capita, population density, or the share of population in manufacturing, at different points in time), δ_s is a state fixed effect, $\mathbf{X}_c$ is a vector of control variables, and ε_c is an error term.

Throughout the paper, I consider different specifications that sequentially expand the set of controls to include state fixed effects, geoclimatic controls, crop-specific

[5] For the years in which labor force data can be constructed using employment micro-data from Ruggles et al. (2020), the correlation between the share of the *labor force* in manufacturing and the share of *population* in manufacturing is above 0.9; the empirical analysis in the following sections yields similar results using either outcome for the years in which both are available.

Table 2—Agricultural Diversity and Long-Run Development: OLS Results

	(1)	(2)	(3)	(4)	(5)
Panel A. Dependent variable: ln Population Density 2000					
Ag. Diversity$_{1860}$	2.994	2.020	2.370	2.206	2.137
	(0.312)	(0.584)	(0.539)	(0.521)	(0.257)
R^2	0.095	0.277	0.354	0.372	0.612
Panel B. Dependent variable: ln Income per Capita 2000					
Ag. Diversity$_{1860}$	0.547	0.241	0.249	0.221	0.269
	(0.0559)	(0.0899)	(0.0828)	(0.0816)	(0.0652)
R^2	0.102	0.297	0.361	0.373	0.478
Panel C. Dependent variable: Share of Population in Manufacturing 1900					
Ag. Diversity$_{1860}$	0.0985	0.0358	0.0404	0.0334	0.0333
	(0.0108)	(0.00949)	(0.00919)	(0.00887)	(0.00825)
R^2	0.087	0.439	0.461	0.468	0.618
State fixed effects	No	Yes	Yes	Yes	Yes
Geoclimatic controls	No	No	Yes	Yes	Yes
Crop-specific controls	No	No	No	Yes	Yes
Socioeconomic conditions in 1860	No	No	No	No	Yes
Observations	1,821	1,821	1,821	1,821	1,821

Notes: See Data Appendix for variable definitions and sources. The means of the dependent variables in panels A, B, and C are 4.37, 10.06, and 0.034, respectively. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

controls, and socioeconomic initial conditions. I discuss these subsets of control variables in detail below.

Table 2 reports OLS estimates of the coefficient on 1860 agricultural diversity in regressions for three different outcome variables: the log of population density in 2000 (panel A), the log of income per capita in 2000 (panel B), and the share of the population employed in manufacturing in 1900 (panel C). Online Appendix A.1 shows that the results are robust to considering indexes of agricultural diversity with alternative functional forms and adjustments that address potential concerns about measurement error in the agricultural production data.

The first column of Table 2 displays the coefficient on agricultural diversity when this is the only regressor, while the second column reports results from regressions that also include state fixed effects. State fixed effects ensure that the estimated coefficient on agricultural diversity does not pick up the effects of other factors that display sharp regional differences like those displayed by agricultural diversity, e.g., between the North and the South. The estimated coefficients are considerably lower when only within-state variation is considered, but they are still significant, positive, and sizable.

In columns 3–5, as I sequentially expand the set of controls, the coefficients on agricultural diversity remain significant and stable in magnitude; the estimates in these columns imply that an increase of 1 standard deviation in agricultural diversity (0.1234) is associated with increases of about 26–29 log points (30–34 percent) in 2000 population density, 3–5 percent in 2000 income per capita, and 4–5 percentage points in the share of population in manufacturing in 1900.

Geoclimatic controls include county area, two measures of potential agricultural productivity, mean annual temperature, rainfall, latitude and longitude, and distance to the ocean or the Great Lakes.[6] Including geoclimatic controls is important, as they may be correlated with agricultural diversity and may also have direct effects on development. Agricultural productivity's impacts on development are the subject of a large literature (see Gollin 2010), while proximity to waterways is a key determinant of access to markets, which in standard trade models increases both specialization and income levels. Section IVC and online Appendix A.2 show that the results are robust to controlling flexibly for these geoclimatic features and additional ones (including several land productivity measures, distance to the fall line, etc.).

Crop-specific controls capture the dominance of specific agricultural products in 1860. Since the diversity index is based on the squared values of individual shares and some agricultural products (wheat, corn, hay, cotton, animals slaughtered) represented large shares of agricultural output, these shares may be strongly correlated with agricultural diversity. Thus, the estimated coefficient for diversity could pick up the positive or negative effects of being specialized in one of those particular products. To address this, I include dummies for each of the five major agricultural products that take a value of 1 when the product has the largest share in a county's agricultural production. I also include a dummy that takes a value of 1 when the combined share of plantation crops other than cotton (tobacco, sugarcane, and rice) is larger than the share of any other individual crop.[7] Online Appendix A.3 shows that the results are robust to controlling for specialization in particular crops in different ways and adding controls for clover, peas, and beans (together with hay, already included in the baseline, these were the crops most commonly used in rotation schemes).

Socioeconomic controls (in all cases measured in 1860) include the urbanization rate; farm output per improved acre (in logs); the shares of the population corresponding to enslaved people, foreigners, people below 15 years of age and people above 65; distance to the nearest railroad; distance to steamboat-navigated rivers; and a measure of "market potential." Market potential in county c is calculated, following the classic definition from Harris (1954), as $\sum_{k \neq c} d_{c,k}^{-1} N_k$, where k is the index spanning other counties, $d_{c,k}$ is the distance between county c and county k, and N_k is the population of county k (here, in 1860).[8] Section IVC shows that the results are robust to including additional socioeconomic controls (in particular, to account more thoroughly for levels of development in 1860) and to excluding from the sample counties that had any urban population in 1860.

[6] As measures of agricultural productivity, I use the max and the mean of 22 product-specific climate-based (normalized) measures of potential yield provided by FAO-GAEZ. The data are explained in detail in Section IVA and the Data Appendix. I normalize each product-specific potential yield by the maximum attained in the sample to make yields for different products comparable.

[7] Specialization in plantation crops is considered to be a crucial determinant of the US North–South divide (Engerman and Sokoloff 2002), and it could also affect within-state variation in development.

[8] Donaldson and Hornbeck (2016) show that Harris's *ad hoc* measure is similar to a first-order approximation to market access derived from an Eaton-Kortum trade model; in the latter, neighboring populations are weighted by the inverse of trade costs elevated to the trade elasticity, for which they use a baseline value of 8.22. Measuring market potential as $Market_c = \sum_{k \neq c} d_{c,k}^{-8.22} N_k$ does not qualitatively affect my findings.

Crop-specific controls and socioeconomic controls (all of them measured in 1860) capture initial conditions that are predetermined with respect to the outcome variables, but they could be endogenous. They are not predetermined with respect to the regressor of interest and thus may be "bad controls." If these variables capture mechanisms through which diversity affects outcomes, including them as controls would mask the true effects. Therefore, the specification that includes state fixed effects and geoclimatic controls but not these initial conditions (column 4 in Table 2) may be preferable. However, including them may be appropriate insofar as there can be correlations between them and agricultural diversity that do not reflect causality from the latter to the former. Thus, the robustness of the results to controlling for those variables is reassuring.

In line with the approach suggested by Bester, Conley, and Hansen (2011) to address spatial serial correlation, standard errors in all specifications are clustered on 60-square-mile grid squares that completely cover the counties in the sample. As shown in online Appendix A.4, the results are qualitatively similar to those obtained with clustering on other levels or with Conley (1999) standard errors for various distance cutoffs.

Next, in Table 3, I report estimates of the association between early agricultural diversity and long-run population density for each of the seven Census divisions in my sample (namely, the New England and Mid-Atlantic divisions in the Northeast region, East North Central and West North Central in the Midwest region, and South Atlantic, East South Central, and West South Central in the South). The results show positive and significant associations for all subsamples except for New England (which might be due to lack of power, as there are only 67 counties in this subsample).

To conclude this preliminary exploration, I examine the statistical association of 1860 agricultural diversity with population density and the share of population in manufacturing at different points in time. I focus on these development outcomes as data are available all the way back to 1860. Figure 2 shows estimates of the coefficient on $Ag.\ Diversity_{1860}$, with the corresponding 95 percent confidence intervals, for the specification that includes state fixed effects and geoclimatic controls. For both development outcomes, the estimated coefficients on 1860 agricultural diversity increase over the Second Industrial Revolution; for the share of population in manufacturing, the association becomes significant in the late nineteenth century, and for population density, in the early twentieth century. Thereafter, the statistical associations remain positive and significant.[9]

IV. Instrumental Variable Strategy and Results

The estimations presented in the previous section show suggestive correlations. However, the positive association between agricultural diversity and development could be driven by omitted variables that induced higher diversity in 1860 and also improved subsequent economic performance. For example, diversity might partly

[9] I run a separate regression for each outcome in each year; the estimates are the same as those obtained from pooling observations from all time periods and letting all regressors have time-varying effects.

Table 3—Agricultural Diversity and Development within Census Divisions

	Northeast		Midwest		South		
	New England	Mid-Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central
Census divisions:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable: ln Population Density 2000							
Ag. Diversity$_{1860}$	1.471	2.203	1.762	1.826	3.347	2.130	2.039
	(1.781)	(0.866)	(0.454)	(0.685)	(0.619)	(0.585)	(0.434)
R^2	0.975	0.899	0.689	0.633	0.516	0.542	0.460
Observations	67	146	393	258	449	300	208
State fixed effects	Y	Y	Y	Y	Y	Y	Y
Geoclimatic controls	Y	Y	Y	Y	Y	Y	Y
Crop-specific controls	Y	Y	Y	Y	Y	Y	Y
Socioeconomic controls	Y	Y	Y	Y	Y	Y	Y

Notes: See Data Appendix for variable definitions and sources. The mean of the dependent variable for the whole sample is 4.37. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

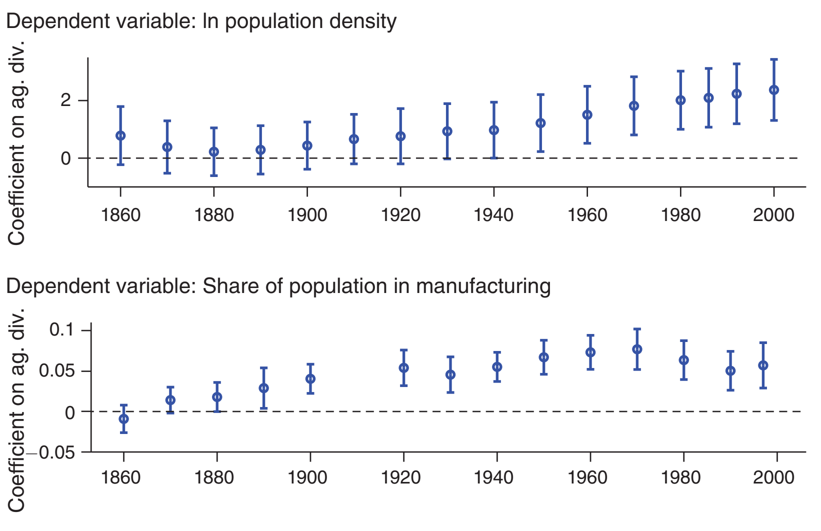

Figure 2. Early Agricultural Diversity and Development

Notes: The figure displays the estimated coefficients on 1860 agricultural diversity from OLS regressions with the log of population density and the share of population in manufacturing at different times as outcomes variables, controlling for state fixed effects and geoclimatic conditions. Also shown are the 95 percent confidence intervals based on robust standard errors clustered on 60-square-mile grid squares. See Data Appendix for variable definitions and sources.

reflect a high propensity to adopt new ideas, the extent of crop rotation, and/or higher levels of human capital, which would themselves boost the process of development. Diversity in agricultural production may also reflect a more diversified local demand driven by higher income in 1860, which could in turn be positively correlated with long-run development.

On the other hand, if there are omitted variables that are negatively correlated with diversity and positively associated with subsequent economic growth, the OLS estimates would be negatively biased. For instance, diversity might partly reflect risk aversion, which could in turn discourage investments in manufacturing insofar as this sector was perceived as highly risky. Diversity might also reflect the prevalence of traditional agriculture (whereby farmers grow most of what they need for their own subsistence), which may in turn be associated with poor economic outcomes. Market access, if not adequately captured by the controls, could also generate a negative bias, since larger potential gains from trade would induce both specialization and growth.

The controls in the baseline OLS estimations and the additional controls included in Section IVC and online Appendix A aim to capture some of the factors mentioned above (e.g., crop rotation, early development, market access). However, the controls may not fully capture these factors. Moreover, other factors are unobservable. Thus, concerns about omitted variable bias persist.

To get closer to identification of causal effects, this section introduces an instrumental variable strategy that exploits exogenous variation in agricultural production patterns generated by climatic features. I start by describing the construction of an instrumental variable for agricultural diversity. I then present and discuss the IV estimates. The last subsection and online Appendix A report a number of identification and robustness checks that lend further credence to my estimates of the effects of agricultural diversity on development.

A. *Potential Agricultural Diversity*

The identification strategy relies on the basic insight that agricultural production patterns are partly induced by climatic features. In particular, agricultural diversity is influenced by the dispersion of product-specific potential yields determined by climate. Intuitively, a county that has similar levels of productivity for many different crops is likely to be more diversified than a county with productivity for one crop much higher than for all others.

The FAO's Global Agro-Ecological Zones (GAEZ) project provides measures of maximum attainable yields (in tons per hectare per year) for different crops based on high spatial resolution climatic data and crop-specific characteristics. These measures of crop-specific potential productivity are based on expert knowledge of climatic features affecting agricultural production processes; they do not rely on a statistical analysis of production patterns observed across the world. In addition, though based on climatic records for 1961–1990, they provide good proxies for historical conditions (see Nunn and Qian 2011). I use attainable yields for rain-fed conditions and intermediate levels of inputs/technology, as these correspond most closely to the context under consideration. Figure 3 displays the county-level average values of potential yields for selected crops (wheat, corn, potato, and tobacco).

To construct an instrumental variable based on crop-specific attainable yields from FAO-GAEZ, I use a fractional multinomial logit framework, which generalizes the fractional logit model to an arbitrary number of choices (for a summary and discussion, see Mullahy 2015).

In the context under consideration, the FML model is specified as a system of equations in which the outcome variables are the shares of each agricultural product i

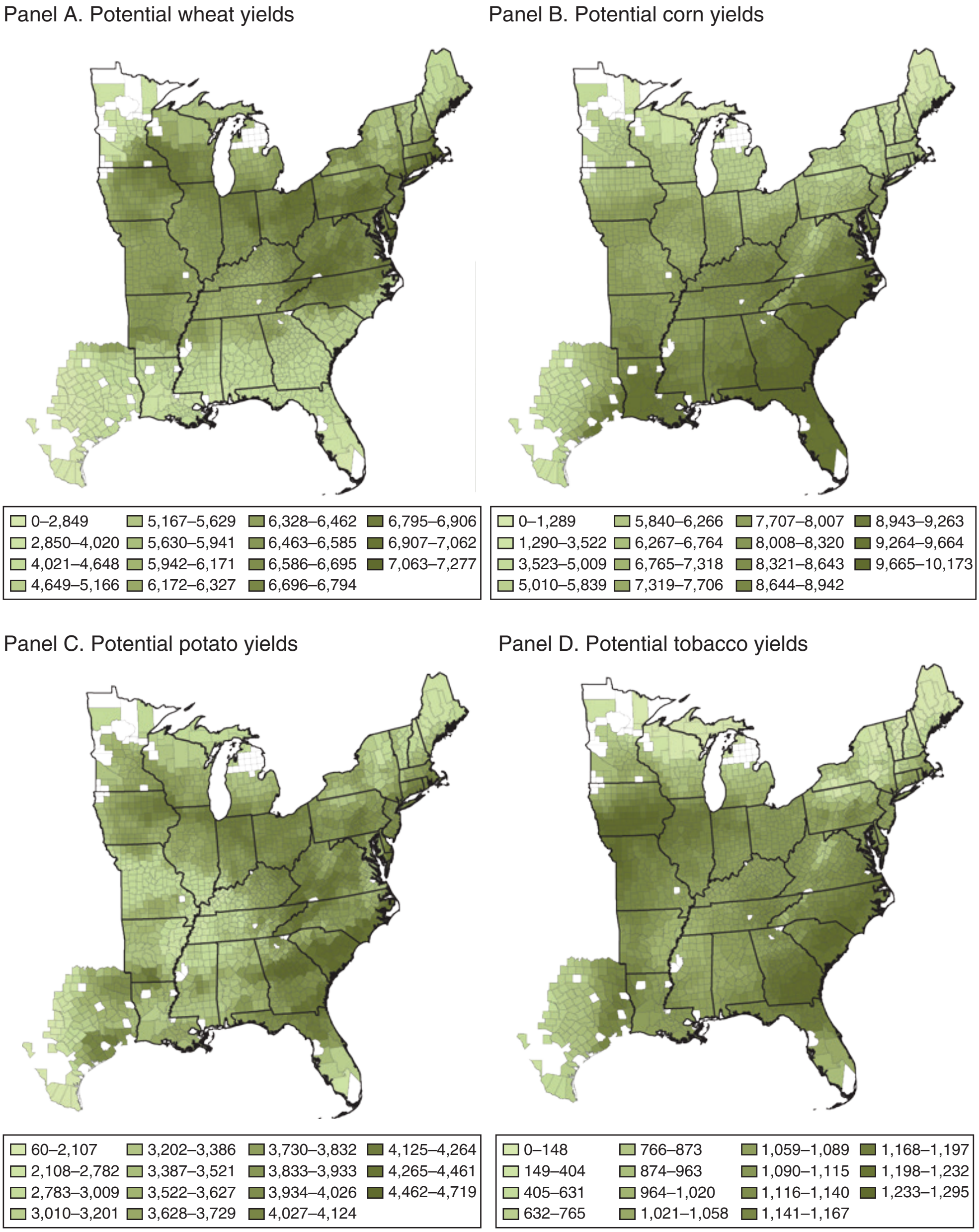

Figure 3. Potential Yields for Selected Crops

Notes: The maps display county-level means of agroclimatic attainable yields from IIASA and FAO (2012) for wheat, corn, potato, and tobacco, in tons per hectare per year, for rain-fed conditions and intermediate levels of inputs/technology. See Data Appendix for variable definitions and sources.

in total agricultural output in county c (i.e., θ_{ic}, for $i = 1, 2, \ldots, 36$) and the regressors are the crop-specific potential yields $\mathbf{A}_c$.[10]

[10] In the estimation of the model, the vector of potential yields includes all relevant crop-specific productivities available from FAO: alfalfa, barley, buckwheat, cane sugar, carrot, cabbage, cotton, flax, maize, oats, onion, pasture

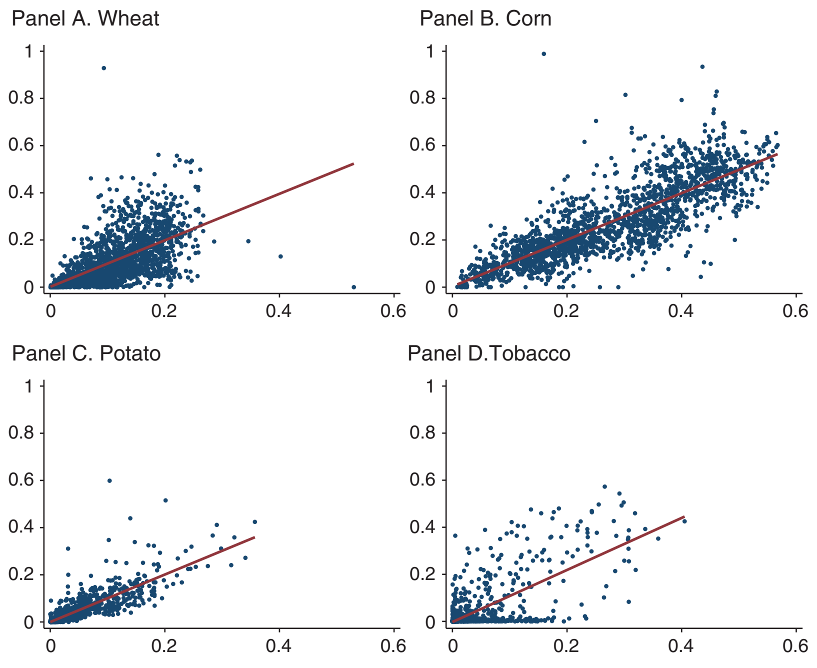

Figure 4. Actual and Predicted Shares of Selected Crops

Note: The figure displays scatterplots of the actual shares (vertical axes) and predicted shares (horizontal axes) of wheat, corn, potato, and tobacco in agricultural production (the predicted shares are obtained from the FML model).

The functional form of the FML model is

$$\hat{\theta}_{ic} = E\left[\theta_{ic} | \mathbf{A}_c\right] = \frac{e^{\beta_i' \mathbf{A}_c}}{1 + \sum_{j=1}^{I-1} e^{\beta_j' \mathbf{A}_c}}. \tag{2}$$

By construction, $\sum_{i=1}^{I} \hat{\theta}_{ic} = 1$, i.e., the predicted shares for each county add up to 1. The parameters are estimated by quasi-maximum-likelihood.

This econometric framework can be motivated by a simple model of optimal crop choice. Recasting the conditional logit framework of choice behavior by McFadden (1974) in terms of profit maximization rather than utility maximization, assume that profits obtained when choosing crop i for a unit of farm resources k are $\pi_{ik} = \beta_i' \mathbf{A}_k + \mu_{ik}$. Farmers are assumed to be price-takers. The estimated parameters reflect the price and cost differentials among agricultural products as well as any other factors that affect profits for different crops. If the error term μ_{ik} is assumed to be i.i.d. with type I extreme value distribution, then choice i is optimal (i.e., $\pi_{ik} \geq \pi_{i'k}$ for all i') with probability $e^{\beta_i' \mathbf{A}_k} / \left(1 + \sum_{j=1}^{I-1} e^{\beta_j' \mathbf{A}_k}\right)$.

grasses, pasture legumes, potato, pulses, rice, rye, sorghum, sweet potato, tobacco, tomato, and wheat. Note that for some of the crops comprised in the agricultural production of US counties, the FAO does not provide measures of attainable yields. The results are very similar if I include measures of overall land productivity in $\mathbf{A}_c$ to serve as proxies for these crops' potential yields. Even without overall productivity measures, the model can provide good predictions for the shares of these crops insofar as their potential yields are correlated with some of the available ones and/or they display production complementarities with specific crops.

Once the coefficients of the FML have been estimated, they are used (in combination with county-level product-specific productivities) to calculate predicted shares for each agricultural product. In all cases, the predicted shares account for a large fraction of the variation in actual shares. Regressing actual on predicted shares for each crop, the coefficients on the latter are always positive and highly significant, even when controlling for state fixed effects and geoclimatic conditions, and the partial R^2 values are large (in most cases, higher than 0.3). Figure 4 plots actual against predicted shares for selected agricultural products.

Using the predicted shares from the FML model for all agricultural products, I calculate an index of potential agricultural diversity. This index is defined by the same formula used for actual diversity but substituting predicted shares for actual shares; that is, $\textit{Potential Ag. Diversity}_c = 1 - \sum_i \hat{\theta}_{ic}^2$. The predicted shares from the FML model are also used to construct a measure of latent agricultural profitability (a weighted average of predicted crop-specific profitabilities with weights given by the predicted shares) included along with several additional controls in Section IVC.

As the predicted shares are good predictors of the actual shares, the measure of potential diversity is a good predictor of actual diversity. Figure 5 shows the scatterplot of the latter against the former. Figure 6 shows the spatial distributions of potential agricultural diversity and its values after partialling out state fixed effects and geoclimatic controls, to give a sense of the variation that is used to identify the effects of agricultural diversity in the IV estimation.

B. *The Effects of Agricultural Diversity on Development*

I estimate the effects of agricultural diversity using potential agricultural diversity as an IV. The identifying assumption is that the measure of potential agricultural diversity, based on the estimation of the FML model, only affects development outcomes through actual agricultural diversity. To alleviate possible concerns about the validity of the exclusion restriction, recall that the FAO measures of potential yields for different crops do not rely on a statistical analysis of observed production patterns. Moreover, note that any determinants of crop choice other than the climate-based productivity measures have their effects loaded onto the residuals of the FML model and thus do not affect the IV estimates of the effects of agricultural diversity.

Despite the considerations above, potential agricultural diversity might still be correlated with geoclimatic features or socioeconomic conditions with direct effects on development outcomes. To address this, the baseline regressions include a wide array of controls further expanded in Section IVC and online Appendix A. The nature of the IV implies a concrete threat of spurious correlations with geoclimatic features. In Section IVC's Table 5 (panel B), I expand the baseline set of geoclimatic controls to include flexible polynomials of several measures of overall land productivity and product-specific attainable yields from FAO; latitude and longitude; temperature and rainfall; and distances to rivers, canals, and the fall line. Section IVC provides a summary of this and other exercises. While the possibility of spurious correlations that make the exclusion restriction invalid can never be ruled out, it

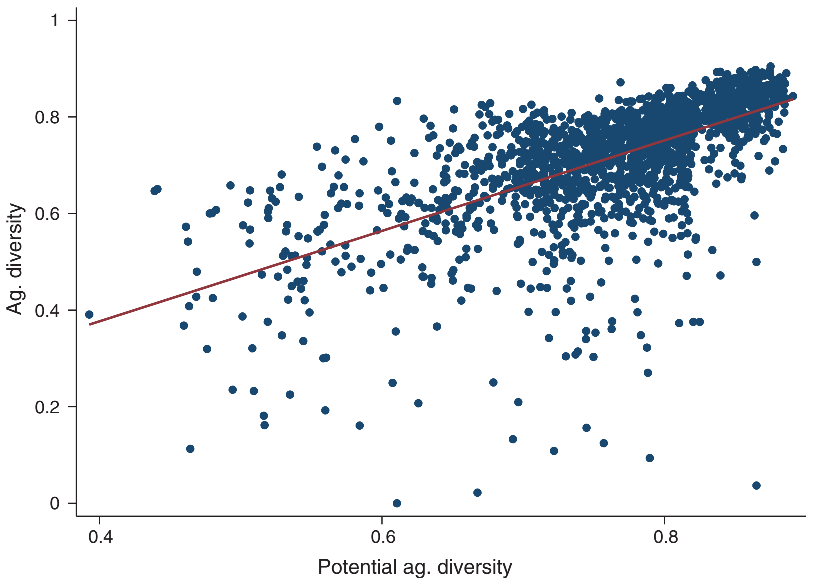

Figure 5. Actual and Potential Agricultural Diversity

Note: The figure displays a scatterplot of actual and potential agricultural diversity (the latter is calculated with the predicted shares from the FML model).

is reassuring that the estimates remain statistically significant and relatively stable throughout a battery of checks.

Table 4 reports the results of the IV estimation using the same specifications considered in the OLS estimation, for the same development outcomes: the log of population density in 2000 (panel A), the log of income per capita in 2000 (panel B), and the share of population in manufacturing in 1900 (panel C). Panel D displays the first-stage estimates and Kleibergen-Paap F-statistics.

For all specifications, the IV estimates indicate significant positive effects of 1860 agricultural diversity on all outcomes, and the IV has strong predictive power. According to the results from my preferred specification (including state fixed effects and geoclimatic controls), reported in column 3, increasing agricultural diversity by 1 standard deviation (0.1234) led to an increase of 0.46 standard deviations in the log of 2000 population density, which amounts to 55 log points, or 73 percent in density levels. By comparison, Bleakley and Lin (2012) estimate that being close to a historical portage site increased county-level population in 2000 by about 77–94 log points (depending on the specification), and Michaels (2011) estimates that oil abundance in the US South led to about 42–102 log points higher 1990 population. Regarding the other outcomes, an increase of 1 standard deviation in agricultural diversity led to increases of 0.29 standard deviations in the log of 2000 income per capita (a 6 percent increase in its level) and 0.25 standard deviations (1 p.p.) in the share of population in manufacturing in 1900.

The observed effects of agricultural diversity on long-run income per capita may be smaller, or larger, than what they would be if labor was immobile across counties.

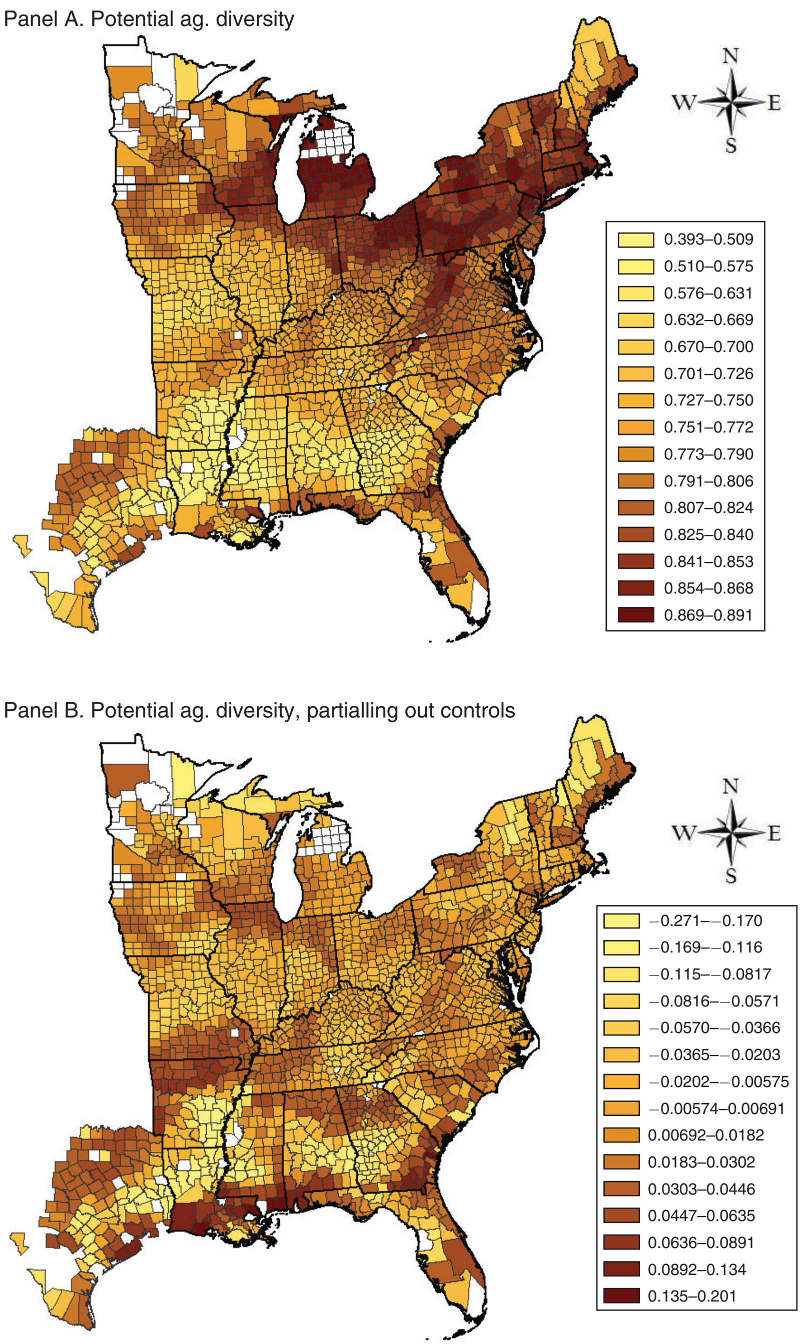

Figure 6. Potential Agricultural Diversity

Note: The maps display the spatial distribution of potential agricultural diversity (calculated with the predicted shares from the FML model) and its values after partialling out state fixed effects and geoclimatic controls.

Table 4—Effects of Agricultural Diversity on Development: IV Estimates

	(1)	(2)	(3)	(4)	(5)
Panel A. Dependent variable: ln Population Density 2000					
Ag. Diversity$_{1860}$	4.539	3.641	4.477	4.276	4.463
	(0.619)	(0.973)	(1.045)	(1.190)	(1.142)
Panel B. Dependent variable: ln Income per Capita 2000					
Ag. Diversity$_{1860}$	0.915	0.449	0.491	0.415	0.439
	(0.0982)	(0.160)	(0.160)	(0.187)	(0.204)
Panel C. Dependent variable: Share of Population in Manufacturing 1900					
Ag. Diversity$_{1860}$	0.169	0.0719	0.0830	0.0639	0.0884
	(0.0220)	(0.0219)	(0.0213)	(0.0251)	(0.0309)
Panel D. First-stage dependent variable: Ag. Diversity$_{1860}$					
Potential Ag. Diversity	0.937	0.624	0.593	0.489	0.398
	(0.0478)	(0.0613)	(0.0638)	(0.0685)	(0.0606)
Kleibergen-Paap Wald F-statistic	384.23	103.43	86.53	51.03	43.06
State fixed effects	N	Y	Y	Y	Y
Geoclimatic controls	N	N	Y	Y	Y
Crop-specific controls	N	N	N	Y	Y
Socioeconomic conditions in 1860	N	N	N	N	Y
Observations	1,821	1,821	1,821	1,821	1,821

Notes: See Data Appendix for variable definitions and sources. The means of the dependent variables in panels A, B, and C are 4.37, 10.06, and 0.034, respectively. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

Without labor flows, productivity gains translate into differences in income per capita, with no effects on population density (except through fertility). In contrast, with relatively high labor mobility, like in the context of US counties, income differentials induce labor flows, and thus productivity gains arising from early agricultural diversity would induce differences in population density. Differentials in income can only remain in equilibrium insofar as they are compensated by differentials in living costs and amenities. As people move to places with higher productivity and initially higher income per capita, congestion pushes up living costs and may also erode some of the initial productivity differential (if there are decreasing returns of some sort) or augment it (through agglomeration forces). In any case, the expected effects of productivity gains on income per capita and population density go in the same direction, but it is important to note that the magnitude of the observed effects on income and population does not reflect only the direct impacts but also the ensuing movements toward spatial equilibrium.

I continue to follow the approach suggested by Bester, Conley, and Hansen (2011) to address spatial serial correlation, clustering standard errors on 60-square-mile grid squares that completely cover the counties in the sample. A sufficient condition for standard errors to be correct when using a generated instrumental variable (here, potential agricultural diversity) is that the expectation of the error term in the estimating equation conditional on the variables used in the IV construction (here, the crop-specific potential yields) is equal to zero (see Wooldridge 2010). This sufficient condition is satisfied insofar as the estimating equation adequately controls for measures of overall land productivity and climatic variables that may have direct effects on development outcomes (online Appendix A.2 shows that the results are robust to flexibly controlling for an expanded set of land productivity measures and geoclimatic conditions in a number of different specifications).

The IV estimates are larger than the OLS estimates. This difference may be due to a negative bias in the OLS estimation generated by omitted variables that display correlations of opposite signs with 1860 agricultural diversity and with subsequent economic performance (e.g., risk aversion, the prevalence of traditional agriculture, or market access, as discussed above). It could also be due to attenuation bias generated by measurement error. Or perhaps the IV captures differentially higher effects of agricultural diversity on development for the counties in my sample where potential agricultural diversity effectively induced actual diversity.

I also examine the effects of early agricultural diversity on development outcomes over time. Estimated effects on (log) income per capita in 1960, 1970, 1980, and 1990 are positive, significant, and similar in magnitude to those for 2000; income per capita data are not available before the mid-twentieth century. The estimated effects on population density and the share of population in manufacturing at different points throughout the 1860–2000 period are displayed in Figure 7 (with the corresponding 95 percent confidence intervals). These are obtained using the regression specification that includes state fixed effects and geoclimatic controls.

The results indicate that early agricultural diversity had positive effects on development outcomes that emerged in the early twentieth century, during the Second Industrial Revolution. In these IV estimates, the effects of agricultural diversity appear to increase in magnitude in the mid-twentieth century. While this may seem surprising, it is consistent with my interpretation of the underlying mechanisms. As discussed in Section V and captured by a model in online Appendix B, early diversity can lead to entry into new activities, new ideas and new skills, then further diversification—a chain of developments plausibly elapsing several decades. Also in line with this interpretation, the observed time pattern may reflect the increasing importance of knowledge-intensive industries, on which early agricultural diversity had differentially positive effects (as established in Section VC).

C. *Identification and Robustness Checks*

I conduct a number of additional exercises that lend further credence to my estimates of the effects of agricultural diversity on development. This section presents some of the results and summarizes the rest, which are reported in detail in online Appendix A.

First, I consider a number of alternative measures of diversity to address possible measurement issues. Next, I perform a battery of checks accounting more thoroughly for initial socioeconomic conditions, flexibly controlling for a wide array of geoclimatic features, including different flexible specifications for distances to waterways and the fall line, and considering various types of controls for specialization in specific crops. The robustness of the estimates lessens concerns about omitted variable bias in the OLS regressions. Moreover, these checks allay concerns

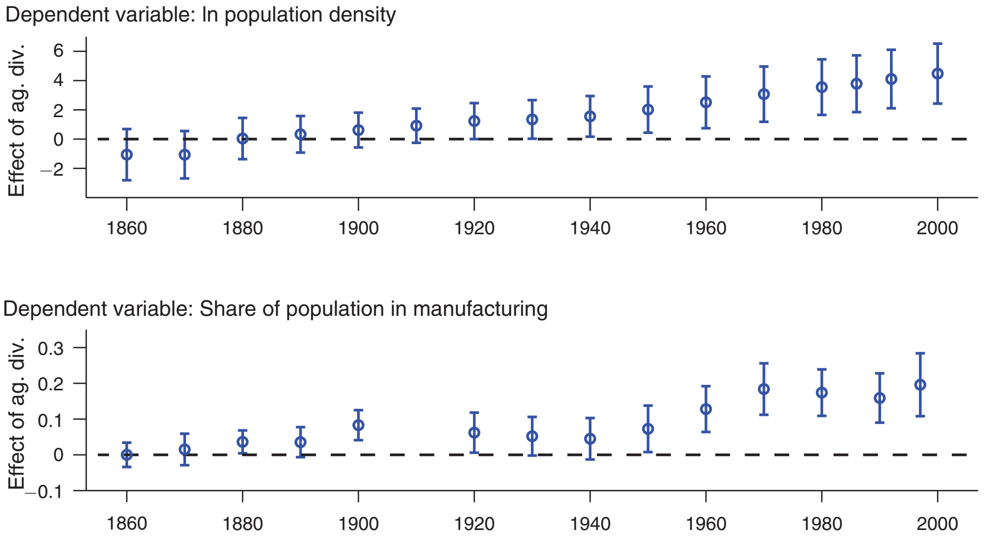

Figure 7. Effects of Agricultural Diversity on Development Outcomes

Notes: The figure displays the estimated coefficients on 1860 agricultural diversity from IV regressions with the log of population density and the share of population in manufacturing at different times as outcomes variables, controlling for state fixed effects and geoclimatic conditions. Also shown are the 95 percent confidence intervals based on robust standard errors clustered on 60-square-mile grid squares. See Data Appendix for variable definitions and sources.

about violation of the exclusion restriction in the IV estimation due to spurious correlations.

Finally, I show that the results are robust to alternative standard errors, and I conduct placebo checks that indicate that the standard errors used in my analysis can effectively curtail spurious correlations driven by spatial autocorrelation.

Alternative Measures of Diversity.—Online Appendix A.1 considers indexes capturing agricultural diversity (or specialization) with different functional forms than the standard Hirschman-Herfindahl index as well as adjustments that address potential concerns about measurement error in the agricultural production data. In all cases, the results are qualitatively the same.

Additional Socioeconomic Controls.—The results are robust to accounting more thoroughly for local economic development in 1860, before the onset of the Second Industrial Revolution. The baseline analysis includes a specification (see, e.g., column 5 in Tables 2 and 4) with various socioeconomic controls measured in 1860: the urbanization rate; farm output per improved acre (in logs); population shares of enslaved persons, foreigners, people below 15 and above 65 years old; distances to railroads and steamboat-navigated rivers; and a measure of "market potential." Columns 1–2 in panel A of Table 5 reproduce the main results controlling for state fixed effects, geoclimatic controls, crop-specific controls, and these baseline socioeconomic controls.

Table 5—Additional Controls for Early Development and Geoclimatic Features

	Dependent variable: ln Population Density 2000					
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)
Panel A. Additional controls for early development						
Ag. Diversity$_{1860}$	2.137	4.463	1.617	4.002	2.208	4.213
	(0.257)	(1.142)	(0.248)	(1.133)	(0.264)	(1.070)
State fixed effects	Y	Y	Y	Y	Y	Y
Geoclimatic controls	Y	Y	Y	Y	Y	Y
Crop-specific controls	Y	Y	Y	Y	Y	Y
Baseline socioeconomic controls 1860	Y	Y	Y	Y	Y	Y
Additional socioeconomic controls 1860	N	N	Y	Y	N	N
Restricted sample	N	N	N	N	Excluding counties with urban pop. 1860 $>$ 0	
R^2	0.612	0.589	0.657	0.634	0.424	0.398
Observations	1,821	1,821	1,821	1,821	1,539	1,539
Panel B. Additional geoclimatic controls						
Ag. Diversity$_{1860}$	2.370	4.477	1.738	4.806	1.713	4.390
	(0.539)	(1.045)	(0.482)	(1.945)	(0.466)	(1.982)
State fixed effects	Y	Y	Y	Y	Y	Y
Baseline geoclimatic controls	Y	Y	Y	Y	Y	Y
Cubic polynomials in:						
Land productivity measures	N	N	Y	Y	Y	Y
Temperature, rainfall, elevation	N	N	Y	Y	Y	Y
Latitude, longitude	N	N	N	N	Y	Y
Distances to rivers, canals, fall line	N	N	N	N	Y	Y
R^2	0.354	0.329	0.493	0.452	0.503	0.473
Observations	1,821	1,821	1,821	1,821	1,821	1,821

Notes: The mean of log population density in 2000 is 4.37. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

Next, I add controls for early industrialization, a proxy for income per capita, human capital, and population density.[11] The results are displayed in columns 3–4 of panel A of Table 5. Columns 5–6 report the results obtained with a restricted sample that excludes all counties (about 15 percent of the baseline sample) that had any urban population in 1860. This exercise addresses the concern that a nonlinear relationship between early urban agglomeration and (potential and/or actual) agricultural diversity, combined with persistence in agglomeration patterns, might be driving the results.

Additional Geoclimatic Controls.—The results are also robust to controlling for additional geoclimatic features and higher-order terms. Potential agricultural diversity, constructed using climate-based measures of crop-specific attainable yields, may be spuriously correlated with geoclimatic features with direct effects on development that are omitted in my baseline specifications. This would violate the

[11] The proxy for income per capita is the sum of agricultural and manufacturing output in per capita terms. As a measure of human capital, I use the literacy rate of white adult males in 1870; the same results are obtained using literacy in 1860, but the sample size is reduced due to missing data for some counties.

exclusion restriction. Panel B of Table 5 reports the coefficient estimates from OLS and IV regressions of 2000 population density (in logs) on 1860 agricultural diversity obtained when I control flexibly for an expanded set of geoclimatic features.

Columns 1–2 of panel B of Table 5 reproduce the main results controlling for state fixed effects and baseline geoclimatic controls. Columns 3–4 report the results obtained when including cubic polynomials in several land productivity measures as well as in temperature, rainfall, and (log) elevation. Besides the baseline measures of land productivity (the max and the mean of 22 product-specific attainable yields from FAO-GAEZ), I now include the first principal component of the FAO-GAEZ measures; the individual FAO-GAEZ measures corresponding to products with shares in output above 1 percent (corn, cotton, alfalfa, pasture grasses, pasture legumes, wheat, oats, potato, sweet potato, tobacco, cane sugar); a measure of latent profitability in agriculture obtained from the FML model (using the estimated coefficients from the model and the FAO attainable yields to compute crop-specific profitability, combined with the predicted product shares); the length of growing period, that is, the number of days during the year when temperature and moisture are conducive to crop growth and development (also from FAO-GAEZ); and an index of overall land suitability for cultivation from Ramankutty et al. (2002). Columns 5–6 report the results obtained when adding cubic polynomials in latitude, longitude, distance to steamboat-navigated rivers, distance to canals, and distance to the fall line. The importance of the latter is underscored by Bleakley and Lin (2012).[12]

Additional Controls for Distances to Waterways and the Fall Line.—Online Appendix A.2 shows that similar estimates of the effects of agricultural diversity on development are obtained from a different flexible specification with dummies corresponding to five bins defined over the range of distances to rivers, to canals, and to the fall line (0–10 miles, 10–30 miles, 30–60 miles, 60–100 miles, and above). Furthermore, I include interactions of these distance dummies, or the cubic polynomials in these distances, with dummies for the Census regions in the sample (Northeast, Midwest, and South).

Additional Controls for Specific Products.—The results are robust to controlling for specialization in particular crops in different ways. The baseline analysis includes crop-specific dummy variables for the five major agricultural products (corn, cotton, animals slaughtered, hay, and wheat) and for plantation crops other than cotton. Online Appendix A.3 shows that the results are robust to controlling for dummies that take a value of 1 when the share of each of those crops (or group of crops) in a county's agricultural output is above 25 percent, for dummies that take a value of 1 when the share is above the seventy-fifth percentile of the distribution of that share in the whole sample, or for the actual shares instead of dummies. This Appendix also adds controls for the presence of clover, peas, and beans; together with hay, these were the crops most commonly used in rotation schemes.

[12] While it seems natural to include distances to steamboat-navigated rivers and canals in this exercise, note that they are not solely determined by geography and climate.

Alternative Standard Errors.—Online Appendix A.4 shows that the results remain robust for various specifications of standard errors aiming to account for spatial autocorrelation. Besides standard errors clustered on 60-square-mile grid squares (my baseline specification), I consider clustering on State Economic Areas, clustering at the state level, and Conley standard errors (computed with the acreg command by Colella et al. 2019) with different distance cutoffs: 25 miles, 50 miles, 100 miles, 200 miles, 300 miles, and 400 miles. The estimated effects of agricultural diversity (displayed in online Appendix Table A.4) are significant for all specifications.

Moreover, I conduct placebo checks to assess whether the standard errors used in my analysis can effectively curtail spurious correlations driven by spatial autocorrelation. In particular, I examine whether spatial noise variables outperform (in terms of significance) agricultural diversity in the main regressions of the paper. Online Appendix A.4 provides full details. The results (displayed in online Appendix Table A.5) suggest that the approaches to inference that I adopt adequately address concerns about spatial autocorrelation.

V. Channels: Diversity, Ideas, and Skills in the Second Industrial Revolution

Having established the significant positive effects of agricultural diversity on long-run development, I proceed to investigate the underlying mechanisms. This section puts forth the hypothesis that these effects are explained by the presence of complementarities and cross-sector spillovers and empirically shows that early agricultural diversity fostered industrial diversification, technological change, and formation of new skills during the Second Industrial Revolution.

A number of theoretical contributions, summarized in Section I, provide relevant insights for my proposed interpretation. If there are complementarities among production inputs, diversity can reduce unit costs. From a dynamic perspective, variety in skills may spur the formation of new skills, insofar as they complement previously existing ones. Moreover, in the presence of cross-sector knowledge spillovers and combinatorial learning (the production of new ideas through combinations of old ideas), diversity can foster technological change. In the context of comparative development across US counties, the relevance of these forces requires significantly high costs of moving goods, skills, or ideas even at a small spatial scale.

The proposed mechanisms can be captured in a formal model, as I do in online Appendix B. Featuring a combination of the ideas mentioned above, the model highlights that the outcomes considered in this section's empirical analysis—industrial diversification, technological change, and skills formation—are interconnected dimensions of the process of structural transformation. With a stylized representation of complementarities and cross-sector spillovers, I show that diversity in skills (initially given by agricultural diversity) favors entry into new activities; this brings new ideas and new skills, which fosters subsequent diversification, boosting the process of growth and structural change.

Highlighting an important aspect of my interpretation, the model portrays growth as driven by structural change, not only from agriculture to manufacturing but also within manufacturing—it is through entry into new industrial activities that the local economy acquires new skills. In addition, the model starkly states a key assumption underlying my explanation of the effects of agricultural diversity across US counties and their long-run persistence: skills and spillovers are highly localized. Furthermore, the model generates an additional testable implication: diversity, beyond increasing industrialization and industrial diversification, favors sectors that require more skills. I examine this prediction in Section VC.

My study of channels focuses on the 1860–1940 period, encompassing the Second Industrial Revolution, over which the effects of early agricultural diversity emerged. The technological transformations of these decades were accompanied by rapid changes in the industrial composition and the occupational structure. There was widespread entry into new industrial activities and rapid growth of skill-intensive sectors. Meanwhile, there was a shift away from traditional occupations and a diversification of labor skills that included the emergence of many new occupations.[13]

A collection of historical examples throughout this section illustrates the variety of skills and technologies required by different agricultural products, their multiple input-output linkages with industrial activities, and the cross-sector spillovers and complementarities underlying the positive effects of diversity on technological change and skill formation. Agricultural diversity may have increased the local diversity of products, ideas, and skills directly, or indirectly, by favoring entry into various agro-processing industries. A neat example of recombinant innovation comes from the history of Ford Motors' assembly line, which integrated insights about continuous flow production from several agro-processing industries.

To empirically examine my explanation of agricultural diversity's long-run effects, I consider a set of key intermediate variables capturing the channels mentioned above. Section VA shows that early agricultural diversity favored subsequent diversification across manufacturing activities (online Appendix A.5 explores the relevance of input-output linkages). In Section VB, I present evidence indicating positive effects of agricultural diversity on patent activity and the share of industrial workers with new skills. Finally, Section VC shows that the shares of industrial workers employed in skill- and knowledge-intensive activities were also positively affected by early agricultural diversity, lending further support to the proposed interpretation.

There are other channels that may explain the effects of agricultural diversity on development. First, agricultural diversity could be associated with agricultural productivity, which could in turn affect the industrialization process. Second, in the presence of product-specific shocks, higher diversity in agriculture would reduce volatility in the value of farm production, which may improve economic performance. Third, agricultural diversity may be inversely related to land concentration, which could in turn affect local institutions (e.g., schools and banks) through political economy mechanisms. My empirical assessment of these channels (briefly summarized in Section VD and presented in detail in online Appendix C), does not support the relevance of any of them.

[13] Between 1880 and 1940, the county-level average number of active manufacturing sectors within my sample increased from 22.3 to 37.9, while the county-level average number of existing occupations in manufacturing increased from 22.8 to 43.3. Between 1860 and 1940, the share of carpenters, blacksmiths, and shoemakers—the 3 most prominent traditional occupations—in national employment across all craftsmen occupations declined from over 44 percent to under 15 percent.

A. *Diversity, from Agriculture to Manufacturing*

Given the presence of multiple linkages between different agricultural products and nonagricultural products, diversity in agriculture may induce diversity in manufacturing. As Hirschman (1977, p. 80) remarked, "development is essentially the record of how one thing leads to another, and the linkages are that record." Loosely speaking, a wider range of (agricultural) things at early stages of development could lead to a more diverse set of (industrial) things later on.

Input-output linkages between agricultural products and industrial sectors in the late-nineteenth-century United States are documented by Kuhlmann (1944) and Page and Walker (1991). Textile production used cotton and wool. The manufacturing of tobacco products was a sizable industrial activity. Flour milling, the nation's largest industrial sector in production value between 1850 and 1880, was the main source of demand for wheat. Distilling industries used rye, barley, and corn to make spirits. Breweries made beer using hops and barley, sometimes other grains too. The growing meat-packing industry and leather industries were supplied by cattle raisers.

In turn, agro-processing industries had multiple forward and backward linkages. Grain processing supplied bakeries, confectionery establishments, and other food industries. Meat-packing establishments produced not only meat products but also leather, lard, candle wax, glue, and fertilizer, which were used in other activities. In addition, different agricultural products and agro-processing had various linkages with producers of tools and machines, packaging supplies, storage equipment, and transportation equipment.

The presence of various agricultural products may favor entry into various manufacturing sectors, which could in turn favor entry into other manufacturing sectors. Insofar as linkages—including not only input-output relationships but also those generated by common labor skills or knowledge spillovers—affect the local structure of production, early agricultural diversity may be conducive to subsequent manufacturing diversity.[14] Online Appendix A.5 shows that a number of industrial activities with known input-output connections with specific agricultural products were more likely to be present the larger the importance of those agricultural products at the local level.

To assess whether early agricultural diversity boosted diversification across industrial activities, I use census employment microdata to construct a measure of diversity across 59 manufacturing sectors (defined in the 1950 classification system of the Census Bureau). For each county c and time period t, I compute the share of each sector s in total industrial employment, $\vartheta_{s,c,t}$, and then use the standard index to measure manufacturing diversity, $1 - \sum_s \vartheta_{s,c,t}^2$.[15]

Figure 8 displays IV estimates of the effects of early agricultural diversity on manufacturing diversity at different points in time between 1860 and 1940, using my

[14] Using recent US data, Ellison, Glaeser, and Kerr (2010) show that input-output relationships, common labor skills, and knowledge spillovers are all important sources of industry co-location.

[15] Complete count data is available for 1880, 1920, 1930, and 1940; for the other years, the measure of industrial diversity is calculated from census samples, which reduces sample sizes (there are 1,010 counties with nonmissing data in 1860, 1,159 in 1870, 1,789 in 1900, and 1,668 in 1910).

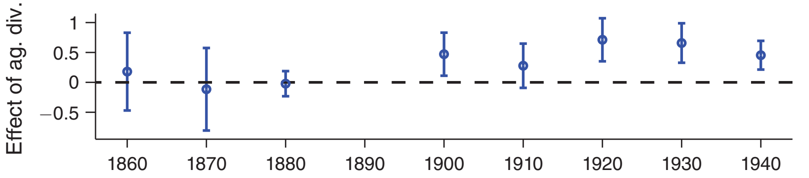

Figure 8. Effects of Agricultural Diversity on Manufacturing Diversity

Notes: The figure displays the estimated coefficients on 1860 agricultural diversity from IV regressions with manufacturing diversity at different times as the outcome variable, controlling for state fixed effects and geoclimatic conditions. Also shown are the 95 percent confidence intervals based on robust standard errors clustered on 60-square-mile grid squares. See Data Appendix for variable definitions and sources. The manufacturing diversity index is constructed from complete count data for 1880, 1920, 1930, and 1940; from 1-in-100 samples for 1860, 1870, 1900, and 1910; and from a 5-in-100 sample in 1900 (in all cases, the largest sample available from Ruggles et al. 2020).

preferred regression specification (controlling for state fixed effects and geoclimatic conditions).[16] The time pattern shows how the effects of early agricultural diversity on manufacturing diversity emerged over the course of the Second Industrial Revolution. An increase of 1 standard deviation in 1860 agricultural diversity led to an increase of 0.48 standard deviations in manufacturing diversity in 1920, when the magnitude of the effects reached a peak.

A local economy with higher diversity has a broader availability of local production inputs. Thus, particularly if transport costs are high, imperfect substitutability between inputs may imply positive effects of agricultural and/or manufacturing diversity on productivity (in the sense of reducing unit costs). Perhaps more relevant, though, are the dynamic implications of economic diversity for the acquisition of new skills and new ideas, to which I now turn.

B. *New Ideas and New Skills*

Diversity can foster entry into new sectors, acquisition of new ideas, and formation of new skills. These are interconnected dimensions of the process of development in a local economy as described by Jacobs (1969, p. 59): "[t]he greater the sheer numbers and varieties of divisions of labor already achieved in an economy, the greater the economy's inherent capacity for adding still more kinds of goods and services." Along the same lines, Jacobs describes development as the process of "adding new work to old work," where the notion of "new work" includes new skills as well as new ideas (encompassing both innovation and imitation).

The positive link between diversity and technological change can be explained by the presence of cross-sector spillovers, recombination, and complementarities. A

[16] The results are similar for other specifications. OLS estimates also produce positive and significant coefficients for 1900, 1920, 1930 and 1940, with lower magnitudes. The results for pre-1920 outcomes have to be interpreted with caution. For 1860, 1870, 1900, and 1910, the lack of complete count census data (see previous footnote) reduces sample size and creates measurement error. Moreover, as emphasized by Katz and Margo (2014), the IPUMS data on sector of employment before 1910 were imputed from data on occupation, which implies an additional source of measurement error.

well-known historical example of cross-product spillovers is the origination of the bow in the bow drill. Classic examples of recombination include windmills (which combine the principles of water mills and sails), the cotton spinning mule (which relied on the moving carriage from Hargreaves's spinning jenny and the rollers of Arkwright's water frame), and incandescent light bulbs (which reinvented candles on the basis of electricity) (Desrochers 2002; Akcigit, Kerr, and Nicholas 2013).

Historical research shows that in nineteenth-century America, agricultural production and its linkages were technologically dynamic and diverse. Different agricultural products were associated with different technologies and skills in production as well as in storage, packaging, transportation, contractual arrangements, and so on. Olmstead and Rhode (2008) offer an impressive account of the rich paths of biological innovation characterizing wheat, corn, cotton, tobacco, livestock, draft animals, and dairy. Two cases of storage and transportation innovations directed to specific agricultural products—the grain elevator and the refrigerated railroad car—became paradigmatic technologies of the nineteenth century (Cronon 1991). Financial derivatives were developed in response to the trading needs of grain and cotton merchants, and equipment leasing was advanced by a food processor seeking financing for capital goods (Glaeser et al. 1992).

Production of agricultural machinery in the mid-nineteenth century was still characterized by small and medium establishments serving local markets and was highly specialized by crop type (Pudup 1987; Page and Walker 1991). For example, corn, wheat, beans, and potatoes all used different types of mechanical planters. While improvements in tools and machines were often specific to one crop or group of crops, the enlargement of producers' know-how presumably yielded cross-product spillovers and higher potential for later recombinations. The agricultural machinery sector also had technological complementarities with other industries. The development of the plow, for instance, relied heavily on advances in iron metallurgy (Pudup 1987).

Agro-processing industries still accounted for a large share of GDP in the late nineteenth century, and they were technologically progressive. In the 1870s and 1880s, they were among the first industries to adopt the new production technologies of the Second Industrial Revolution (Chandler 1977). The diversity of ideas acquired from the application of continuous-flow methods in different agro-processing industries was absorbed in a salient historical case of recombination—the famed Ford Motors' assembly line, which incorporated insights from flour mills, meat-packing establishments, breweries, and canning factories (Hounshell 1985). This collection of historical examples and remarks suggests that agricultural diversity may have enhanced cross-fertilization and recombinant innovations.

In my proposed interpretation, the adoption of new ideas and the formation of new skills are interlinked dimensions of the process of structural change fostered by initial diversity. Following the notion of "new work" from Jacobs (1969), Lin (2011) interprets the presence of workers in new occupations as a measure of technological change and related changes in labor markets. One of his findings (using US data for recent decades) is that new work is more prevalent in locations with high initial levels of industrial diversity. The dynamic effects of diversity under skill complementarities are emphasized by Hausmann and Hidalgo (2011) and in the model proposed in online Appendix B. Diversity, insofar as it reflects a wider range of available skills, increases the return to acquiring new skills that complement the previously existing ones.

To assess the effects of early agricultural diversity on the acquisition of new ideas, I consider a proxy for technological dynamism over the Second Industrial Revolution: the county-level average number of patents per capita per decade between 1860 and 1940.[17] I interpret patent activity as a measure of local technological dynamism broadly defined since it captures not only new frontier technologies but also micro-innovations adapting technologies to local conditions as well as unsuccessful attempts. I also consider the average number of technological classes recorded per patent, calculated with the same patent data. This measures the diversity of technological activity at the local level and could be interpreted as a proxy for recombinant innovation.

To assess how diversity affected the formation of new skills, I focus on skills that emerged during the Second Industrial Revolution. Using census employment data, I identify the high- and middle-skill occupations (as classified by Katz and Margo 2014) that appear in the manufacturing sector in 1940 but were nonexistent in the 1860 IPUMS census sample, such as electricians, toolmakers, and auto mechanics (see Data Appendix for details). I then use the share of these occupations in industrial employment in 1940, a measure of the presence of new skills that resembles the notion of "new work" in Jacobs (1969) and Lin (2011).

I report OLS and IV estimates for three specifications, controlling for (i) state fixed effects, (ii) state fixed effects and geoclimatic conditions, and (iii) state fixed effects, geoclimatic conditions, crop-specific controls, and socioeconomic conditions. The estimates of the effects of early agricultural diversity on the log of average patents per capita per decade in 1860–1940, the log of the average number of technological classes per patent in the same period, and the share of manufacturing employment in new occupations in 1940 are reported in panels A, B, and C of Table 6, respectively.[18]

The results in all specifications indicate positive significant effects of agricultural diversity on the acquisition of new ideas, technological diversity, and new skills. According to the IV estimates in my preferred specification (column 4), a 1 standard deviation increase in early agricultural diversity led to increases of 26 percent in patents per capita per decade, 4 percent in the average number of technological classes per patent, and 0.6 p.p. in the share of new skills in manufacturing employment.

C. *Knowledge- and Skill-Intensive Industrial Activities*

The mechanisms emphasized above have implications about the effects of early agricultural diversity across different industrial activities. If the positive effects of diversity are explained by the role of knowledge spillovers and skill complementarities in the acquisition of new ideas and new skills, then we would expect the effects

[17] I divide patents in each decade by initial population and take the average across decades. The patent data come from Petralia, Balland, and Rigby (2016).

[18] The sample in panels A and B is reduced to 1,811 counties as 10 counties with no patents drop out. The sample in panel C is reduced to 1,818 as 3 counties with no manufacturing workers recorded in 1940 drop out.

Table 6—Effects of Agricultural Diversity on New Ideas and New Skills

	Specification 1		Specification 2		Specification 3	
	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: ln(Patents per capita per decade, 1860–1940)						
Ag. Diversity$_{1860}$	0.894	2.158	0.881	1.873	0.862	2.120
	(0.307)	(0.723)	(0.307)	(0.713)	(0.279)	(1.016)
R^2	0.525	0.516	0.547	0.542	0.633	0.626
Observations	1,811	1,811	1,811	1,811	1,811	1,811
Panel B. Dependent variable: ln(Technological Classes per Patent, 1860–1940)						
Ag. Diversity$_{1860}$	0.0634	0.240	0.101	0.289	0.125	0.408
	(0.0453)	(0.122)	(0.0432)	(0.135)	(0.0437)	(0.191)
R^2	0.085	0.074	0.109	0.097	0.131	0.111
Observations	1,811	1,811	1,811	1,811	1,811	1,811
Panel C. Dependent variable: Share of New Skills in Manufacturing Employment, 1940						
Ag. Diversity$_{1860}$	0.0201	0.0535	0.0212	0.0492	0.0199	0.0657
	(0.00660)	(0.0182)	(0.00684)	(0.0196)	(0.00677)	(0.0306)
R^2	0.140	0.126	0.149	0.140	0.182	0.164
Observations	1,818	1,818	1,818	1,818	1,818	1,818
State fixed effects	Y	Y	Y	Y	Y	Y
Geoclimatic controls	Y	Y	Y	Y	Y	Y
Crop-specific controls	N	N	N	N	Y	Y
Socioeconomic controls	N	N	N	N	Y	Y

Notes: See Data Appendix for variable definitions and sources. The means of the dependent variables in panels A, B, and C are −5.51, 0.68, and 0.023, respectively. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

of diversity to be larger in activities where these forces are more prevalent, namely in knowledge- and skill-intensive activities.

A differentially positive association between economic diversity and high-tech or new activities is hinted at by the urban and regional economics literature (e.g., by the empirical studies of Henderson, Kuncoro, and Turner 1995 and Neffke et al. 2011 and the theoretical contributions of Duranton and Puga 2001 and Helsley and Strange 2002). In the model presented in online Appendix B, the same features that generate positive effects of diversity on industrialization and long-run development—complementarities and cross-sector spillovers—also generate a differentially positive impact on industrial activities with high levels of complexity.

To assess whether early agricultural diversity was conducive to the development of knowledge- and skill-intensive industrial activities, I consider the shares of each of these two types of activities in overall manufacturing employment in 1940, in the aftermath of the Second Industrial Revolution. I define as knowledge-intensive industries those that had above-median percentages of engineers and scientists in total industry employment, and as skill-intensive those that had above-median educational attainment of workers. The correlation between the shares of these two types of activities across counties in the sample is 0.58.

Table 7 presents estimates of the effects of 1860 agricultural diversity on the shares of manufacturing workers in knowledge-intensive activities (panel A)

Table 7—Effects of Agricultural Diversity on Knowledge- and Skill-Intensive Activities

	Specification 1		Specification 2		Specification 3	
	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable: Share of Knowledge-Intensive Activities in Manufacturing, 1940						
$Ag.\ Diversity_{1860}$	0.190	0.559	0.223	0.551	0.227	0.579
	(0.0518)	(0.167)	(0.0493)	(0.161)	(0.0532)	(0.239)
R^2	0.403	0.378	0.438	0.420	0.473	0.457
Observations	1,818	1,818	1,818	1,818	1,818	1,818
Panel B. Dependent variable: Share of Skill-Intensive Activities in Manufacturing, 1940						
$Ag.\ Diversity_{1860}$	0.0968	0.488	0.156	0.533	0.144	0.622
	(0.0518)	(0.172)	(0.0505)	(0.164)	(0.0550)	(0.248)
R^2	0.339	0.309	0.388	0.362	0.424	0.392
Observations	1,818	1,818	1,818	1,818	1,818	1,818
State fixed effects	N	N	Y	Y	Y	Y
Geoclimatic controls	N	N	Y	Y	Y	Y
Crop-specific controls	N	N	N	N	Y	Y
Socioeconomic controls	N	N	N	N	Y	Y

Notes: See Data Appendix for variable definitions and sources. The means of the dependent variables in panels A and B are 0.38 and 0.39, respectively. Robust standard errors clustered on 60-square-mile grid squares are reported in parentheses.

and skill-intensive activities (panel B) in 1940. For all specifications, the results indicate positive and significant effects. According to my preferred specification (column 4), increasing 1860 agricultural diversity by 1 standard deviation would lead to increases in the shares of knowledge- and skill-intensive activities in total manufacturing employment of 6.8 and 6.6 p.p., respectively.

These results show that agricultural diversity had differentially positive effects on activities with high levels of complexity, lending further support to the hypothesis that diversity's long-run impact is explained by the presence of complementarities and cross-sector spillovers.

D. *Other Channels*

There are other channels, unrelated to the types of complementarities and cross-sector spillovers discussed above, that could account for the positive effects of agricultural diversity on development. First, agricultural diversity could lead to changes in agricultural productivity that could in turn boost industrialization. Second, by reducing exposure to product-specific shocks, diversification in agricultural production would decrease volatility and perhaps foster economic performance. Third, agricultural diversity may have been associated with lower land concentration, possibly affecting local institutions. This section briefly summarizes my empirical assessment of these channels, presented in detail in online Appendix C.

To examine whether agricultural diversity affected development through agricultural productivity, I estimate the effect of the agricultural diversity on land productivity as well as the effect of the latter on industrialization. The estimated effects (reported in online Appendix Table C.1) are not significant for either of the two links.

Next, I examine whether agricultural diversity favored development by reducing volatility in the value of agricultural output. I construct a measure of predicted volatility based on the initial county-level production mix and the year-to-year evolution of national prices. Then, I compare the estimated effects of early agricultural diversity on industrialization obtained with and without controlling for this measure. If diversity positively affected industrialization by reducing volatility, this control should have a negative and significant coefficient, and its inclusion would reduce the coefficient on diversity. However, the evidence (displayed in online Appendix Table C.2) is not in line with these predictions.

Finally, I assess whether agricultural diversity affected development by shaping land concentration and thus activating political economy mechanisms. To do this, I empirically study the relationship between agricultural diversity and land concentration as well as the effects of agricultural diversity on key institutions that displayed significant variation at the local level—schools and banks. The results (reported in online Appendix Tables C.3 and C.4) do not support the relevance of this channel.

VI. Conclusion

This paper shows that early agricultural diversity had positive and persistent effects on development across US counties. My identification strategy relies on climate-based measures of crop-specific potential yields, which I use to construct an index of potential agricultural diversity. Using this index as an IV, I find that a 1 standard deviation increase in 1860 agricultural diversity led to increases in 2000 levels of population density and income per capita of about 73 percent and 6 percent, respectively. The positive effects of early agricultural diversity emerged during the Second Industrial Revolution, when it boosted the process of structural change.

I advance the hypothesis that the positive effects of agricultural diversity are explained by the presence of complementarities and cross-sector spillovers. As illustrated by a collection of historical examples from the American Second Industrial Revolution and captured formally in the model presented in online Appendix B, agricultural diversity can foster entry into new activities and the acquisition of new ideas and new skills. Providing support for this interpretation, I show that early agricultural diversity led to higher levels of industrial diversification, patent activity, and employment in new occupations. Moreover, consistent with an additional implication of the model, agricultural diversity positively affected the shares of industrial workers employed in knowledge- and skill-intensive sectors. In contrast, as shown in online Appendix C, the evidence does not support the relevance of other possible channels.

My findings add to a growing body of evidence about deeply rooted factors that, although no longer directly relevant, have shaped historical paths of economic growth and thus display persistent effects on comparative development. My focus on the role of agricultural diversity sheds new light on the role of agriculture in the process of development and also on the relevance of diversity's effects on growth through the acquisition of skills and ideas. Given that these effects cannot be captured in standard two- or three-sector models of structural change, the paper suggests that models with finer levels of aggregation can yield new insights about the development process.

The findings from the history of US counties going back to 1860 may be relevant for understanding the contemporary growth trajectories of developing countries, though naturally this cannot be taken for granted. To further our understanding of the role of diversity, and before any policy implications can be firmly established, future research should attempt to pin down as precisely as possible the mechanisms through which diversity affects development in different contexts across space and time.

Data Appendix

A. *Variable Definitions and Sources*

Outcome Variables.—

Population Density: Population/area. Digitized US Census data on population for 1860–1940 are drawn from Manson et al. (2019). All the data and documentation from this source can be found at http://doi.org/10.18128/D050.V14.0. Data for 1940 onward are drawn from digitized County Data Books compiled by the US Census Bureau, available from Haines and ICPSR (2010). All the data and documentation from this source can be found at https://doi.org/10.3886/ICPSR02896. For 2000, I use the data from the Bureau of Economic Analysis (2015). Data can be downloaded from https://apps.bea.gov/iTable/index_regional.cfm by selecting Table CAINC1 Personal Income, Population, Per Capita Income. The data on area are the same I use as a geoclimatic control (see below).

Income Per Capita: Total personal income/population. Data on personal income and population in 2000 are obtained from the Bureau of Economic Analysis (2015).

Share of Population in Manufacturing: Average yearly number of manufacturing workers over total population. Digitized US Census data for 1860–1940 are drawn from Manson et al. (2019). Data for 1940 onward are drawn from digitized County Data Books compiled by the US Census Bureau, available from Haines and ICPSR (2010).

Manufacturing Diversity: Defined as 1 minus the Herfindahl index of manufacturing employment shares of 59 sectors in total manufacturing employment. Calculated with complete count employment census micro-data for 1880, 1920, 1930, and 1940; 1-in-100 samples for 1860, 1870, 1900, and 1910; and a 5-in-100 sample for 1900, in all cases from Ruggles et al. (2020). The variable recording the sector in the micro-data is IND1950; the manufacturing sectors are the ones with codes between 306 and 499. All the data and documentation from Ruggles et al. (2020) can be found at http://doi.org/10.18128/D050.V14.0.

Patents Per Capita per Decade, 1860–1940: County-level number of patents divided by initial population in each decade, averaged across decades, between 1860 and 1940. I use data from Petralia, Balland, and Rigby (2016) for 1920–1940.

Average Number of Technological Classes per Patent, **1860–1940:** Using data from Petralia, Balland, and Rigby (2016) for 1860–1940, I calculate the average number of technological classes recorded for all patents in a county.

Share of New Skills in Manufacturing Employment, **1940:** Proportion of a county's manufacturing workers in 1940 who are employed in high- and middle-skill occupations (as classified by Katz and Margo 2014) that had 0 respondents in the 1860 census 1-in-100 sample. Employment data for 1860 (a 1-in-100 sample) and 1940 (complete count) are drawn from Ruggles et al. (2020). The variable recording occupations in the micro-data is *OCC1950*. Among high- and middle-skill occupational categories with 0 respondents in the 1860 census sample, there are 14 that are present in manufacturing in 1940 for at least one county in my sample: chemical engineers; electrical engineers; testing technicians; purchasing agents and buyers; advertising agents and salesmen; office machine operators; cranemen, derrickmen, and hoistmen; electricians; heat treaters, annealers, temperers; airplane mechanics and repairmen; automobile mechanics and repairmen; railroad and car shop mechanics and repairmen; motion picture projectionists; professional nurses (the last two with only a handful of occurrences).

Share of Skill-Intensive Activities in Manufacturing, **1940:** Proportion of a county's manufacturing workers in 1940 who are employed in industrial activities with above-median skill intensity. A sector's skill intensity is the average educational attainment of all workers employed in the sector in the complete count 1940 census data available from Ruggles et al. (2020). Calculations rely on variables *IND1950* and *OCC1950*, which record sectors and occupations, respectively.

Share of Knowledge-Intensive Activities in Manufacturing, **1940:** Proportion of a county's manufacturing workers in 1940 who are employed in industrial activities with above-median knowledge intensity. A sector's knowledge intensity is the percentage of engineers and scientists among all workers employed in the sector in the complete count 1940 census data available from Ruggles et al. (2020). Calculations rely on variables *IND1950* and *OCC1950*, which record sectors and occupations, respectively.

Farm Productivity, **1870:** Farm Output/Improved Acres. Farm output is measured in value terms using county-level data from the census and state-level prices. Data source: Digitized US Census data drawn from Manson et al. (2019).

Bank Density: Number of banks per 1,000 inhabitants. FDIC data on banks drawn from NHGIS (Manson et al. 2019).

Share of Farmland in Top 10 Percent Largest Farms: Share of county-level farmland corresponding to the top 10 percent largest farms. Source: Digitized US Census data from NHGIS (Manson et al. 2019).

School Expenditures Per Capita, **1890:** Ordinary expenditures on public common schools over total population. Data can be found at https://www.aeaweb.org/articles?id=10.1257/000282803322655482. Expenditures data, drawn from Rhode and Strumpf (2003), come from the Eleventh Census of the United States, Vol. 25, Report on Wealth, Debt, and Taxation, Pt. II Valuation and Taxation.

Geoclimatic Controls.—

Land Productivity Measures: Maximum and average of normalized attainable yields for alfalfa, barley, buckwheat, cane sugar, carrot, cabbage, cotton, flax, maize, oats, onion, pasture grasses, pasture legumes, potato, pulses, rice, rye, sorghum, sweet potato, tobacco, tomato, and wheat. I normalize each product's values dividing them by the maximum value for that product in the sample. Measures of attainable yields were constructed by the FAO's Global Agro-Ecological Zones project v3.0 (International Institute for Applied Systems Analysis (IIASA), and Food and Agriculture Organization of the United Nations (FAO) 2012) using climatic data, including precipitation, temperature, wind speed, sunshine hours and relative humidity (based on which they determine thermal and moisture regimes), together with crop-specific measures of cycle length (i.e., days from sowing to harvest), thermal suitability, water requirements, and growth and development parameters (harvest index, maximum leaf area index, maximum rate of photosynthesis, etc). Combining these data, the GAEZ model determines the maximum attainable yield (measured in tons per hectare per year) for each crop in each grid cell of 0.083×0.083 degrees. I use FAO's measures of agroclimatic yields (based solely on climate, not on soil conditions) for intermediate levels of inputs/technology and rain-fed conditions. See the model documentation by Fischer et al. (2012) for additional details. All the FAO data can be found at http://www.fao.org/nr/gaez/about-data-portal/en/. For each crop, I take the raster file on attainable yields and calculate the mean value for each US county in the sample using geographic shapefiles of county boundaries from NHGIS (Manson et al. 2019) and a zonal statistics tool in GIS software. In Section IVC, I consider additional measures of land productivity: the first principal component of the 22 crop-specific attainable yields from FAO-GAEZ; *length of growing period*, the number of days during the year when temperature and moisture are conducive to crop growth and development, also from FAO-GAEZ; individual attainable yields (normalized by the maximum attained in the sample) for products with shares in output above 1 percent; a measure of *latent profitability in agriculture*, obtained from the FML model using the estimated coefficients from the model and the FAO attainable yields to compute crop-specific profitabilities, and the predicted product shares to calculate a weighted average of these profitabilities; and *land suitability for cultivation*, the county-level average of an index of land suitability for cultivation (to be interpreted as the probability that a given area is cultivated) from Ramankutty et al. (2002).

Area: Surface area in square miles, calculated with county shapefiles from NHGIS (Manson et al. 2019) using GIS software.

Temperature: County-level mean annual temperature measured in degrees Celsius. Data source: IIASA and FAO (2012).

Rainfall: County-level average annual precipitation measured in millimeters. Data source: IIASA and FAO (2012).

Elevation: County-level average terrain elevation in kilometers (km). Data source: IIASA and FAO (2012).

Latitude: Absolute latitudinal distance from the equator in decimal degrees, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019).

Longitude: Absolute longitudinal distance from the Greenwhich meridian in decimal degrees, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019).

Distance to the Coastline or Great Lakes: Minimum distance to a point in the coastline or the Great Lakes in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019).

Distance to the Fall Line: Minimum distance to a point in the fall line in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019).

Crop-Specific Controls.—

Dummies for Dominance of Specific Products: Dummy variables for each of the five major products (corn, cotton, animals slaughtered, hay, wheat) and one for plantation crops other than cotton (sugar, rice, and tobacco, combined) that take a value of 1 when the product (or group of products) has the largest share in a county's agricultural production value. Data sources: Agricultural production data from the 1860 Census of Agriculture and state-level prices from Atack and Bateman (1987) and Craig (1993) (both available from Manson et al. 2019).

Socioeconomic Controls.—

Urbanization Rate: Urban population / Total Population. Data source: Digitized US Census data drawn from Manson et al. (2019).

Farm Productivity, 1860: Farm Output/Improved Acres. Farm output is measured in value terms using county-level data from the census and state-level prices. Data source: Digitized US Census data drawn from Manson et al. (2019).

Share of Enslaved People in the Population: Enslaved Persons/Total Population. Data source: Digitized US Census data drawn from NHGIS (Manson et al. 2019).

Share of Foreigners in the Population: Foreign Population/Total Population. Data source: Digitized US Census data drawn from NHGIS (Manson et al. 2019).

Share of the Population Below 15 Years: Population below 15 years/Total Population. Data source: Digitized US Census data drawn from NHGIS (Manson et al. 2019).

Share of the Population 65+ Years: Population 65+ years/Total Population. Data source: Digitized US Census data drawn from NHGIS (Manson et al. 2019).

Distance to Railroads: Minimum distance to a point in a pre-1860 railroad line digitized by Atack (2015c) in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019). The GIS database of railroads can be found at https://my.vanderbilt.edu/jeremyatack/data-downloads/.

Distance to Steamboat-Navigated Rivers: Minimum distance to a point in a steamboat-navigated river digitized by Atack (2015a) in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019). The GIS database of rivers can be found at https://my.vanderbilt.edu/jeremyatack/data-downloads/.

Distance to Canals: Minimum distance to a point in a canal digitized by Atack (2015b) in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019). The GIS database of canals can be found at https://my.vanderbilt.edu/jeremyatack/data-downloads/.

Market Potential: Following the classic definition of Harris (1954), market potential in county c is given by $Market_c = \sum_{k \neq c} d_{c,k}^{-1} N_k$, where k is the index spanning neighboring counties, $d_{c,k}$ is the distance between county c and county k, and N_k is the population of county k (here, in 1860). Distances are calculated from the centroid of each county using GIS software and county shapefiles from NHGIS (Manson et al. 2019). Population data are drawn from NHGIS (Manson et al. 2019).

Agricultural Output Predicted Price Index, Average Rate of Change and Volatility.—In online Appendix C.2, I incorporate two additional control variables: the average and the standard deviation of $\hat{P}_{ct}^{a}$, the rate of change of P_{ct}^{a}, a predicted price index for a county's agricultural production. This index is defined as $P_{ct}^{a} = \sum_i \theta_{ic} p_{it}$, where the θ_{ic} terms reflect the county's agricultural production mix in 1860 and p_{it} is the national price of product i at time t (with p_{i1867} normalized to 1 for all i). After calculating the rate of change $\hat{P}_{ct}^{a} = \left(P_{ct}^{a} - P_{ct-1}^{a}\right)/P_{ct-1}$ for each year, I compute its average and standard deviation over the 1867–1900 period, $Avg.\,\hat{P}^a_{ct}$ and $\mathrm{std}\left(\hat{P}^a_{ct}\right)$. Annual price data from 1867 to 1900 are available for 15 products that account for over 93.5 percent of total agricultural output in 1860 in my sample (animals, barley, buckwheat, butter, cheese, cotton, corn, hay, oats, potatoes, rye, sweet potatoes, tobacco, wheat, and wool). Only products with available price data are considered in the calculation of P^a_{ct}; the θ_{ic} values are computed with the restricted set of products so that they add up to 1. Prices for all products except cotton are drawn from Olmstead and Rhode (2006). For animals, I use the average of the prices of cattle and hogs; the former are available from 1867 onward (a year later than prices of other products), which sets the starting point of my measures of P^a_{ct}. I use the price of butter as a proxy for the price of cheese since the latter is only available (for Wisconsin cheddar) from 1879 onward; the correlation between these prices from 1879 to 1900 is 0.73. For cotton, I use prices in Alabama from Cooley, DeCanio, and Matthews (1977) (I am grateful to Paul Rhode for sharing a copy of this dataset).

B. *Adjustment for Changes in County Boundaries*

Among the counties in the sample, with their boundaries defined in 1860, over 70 percent did not experience subsequent changes in boundaries. Among the remaining counties, over two-thirds have overlaps of over 80 percent in terms of area with a county defined in 2000. In this sense, boundaries were relatively stable for counties in the sample. Still, boundary changes need to be taken into account to have consistent units of observation. To adjust data from a period t to 1860 county boundaries, I use geographic shapefiles of county boundaries from NHGIS (Manson et al. 2019). With the help of GIS software, I determine the county defined in 1860 that contained each county or county fragment defined in period t, discarding all fragments with less than one square mile. Then I calculate the values of period t variables corresponding to counties defined in 1860 by assigning values from period t variables in proportion to the area of the county defined in period t that each county defined in 1860 represents. This procedure would be fully accurate if the quantities measured in county-level aggregates were uniformly distributed over space.

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